

Financial Strategy Library Compression under Partial Information: Resource Constraints and Generative Value

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Abstract

Growing financial strategy libraries consume storage, validation, maintenance, monitoring, and governance resources. Yet a retained strategy can be both an operating asset, contributing to current payoff, and a generative asset, carrying modules needed to produce future descendants. We study a finite partial-information process whose raw library dynamics project onto a sufficient productive state comprising the operating frontier and the generative closure. This reduction defines exact, source-relative compression as the minimum-resource sublibrary preserving both objects, and therefore productive value and available optimal actions. Rechecking frontier and closure after every deletion certifies a safe stepwise path, but an irreducible endpoint need not be globally resource-minimal. The distinction is economically sharp: a deletion that preserves every current payoff can eliminate the complete attainable delayed descendant margin in the sharp bridge construction. Beyond exact preservation, selection under a hard capacity or resource penalty operates over finitely many library value–burden points. Capacity value is stepwise and can have increasing unit increments, while penalized value forms a convex piecewise-affine envelope with discrete changes in the selected library. Exact finite fixtures, registered synthetic stress tests, and retrospective financial audits examine these mechanisms. The audits use held-out descendant quality only after compression and support no causal, forecasting, alpha, or deployable performance claim.

1 Introduction

Financial strategy research is cumulative. New signals, transformations, execution rules, and risk controls produce candidate strategies; candidates that survive verification enter a library and may contribute components to later research. A successful research program therefore creates a growing stock of executable policies, documented procedures, and reusable modules. Growth is productive because it expands both the menu that can be operated today and the material from which future strategies can be constructed.

Retention is not free. Every strategy must be stored, indexed, validated, maintained, monitored, retrieved, documented, and governed. These activities consume computing capacity and scarce human attention even when the strategy is not currently selected. As libraries expand, the retention problem becomes an economic decision rather than a housekeeping exercise: which strategies can be removed, and which apparent redundancies are worth paying to keep?

This paper is organized around one question:

How much of a growing strategy library can be removed while preserving both what the library can operate now and what it can help generate later?

The question separates two roles that are easy to conflate. A strategy is an operating asset when its payoff profile improves the action available at a current belief. It is a generative asset when it carries a signal, transformation, execution component, risk control, or other module needed to produce and verify a descendant. A strategy can matter through either role, through both, or through neither.

Conventional current-performance pruning is tempting because the operating role is directly visible. One can retain enough strategies to reproduce the best current payoff at every belief and delete strategies that never attain that envelope. Such a rule can be entirely correct as a statement about the current action menu. It is incomplete as a statement about a research library. A strategy that is dominated at every current belief may still be the only carrier of a module used by a future project. Removing it changes what can be generated even though it changes no payoff available today. Current redundancy and generative redundancy are therefore different properties.

The distinction leads to a compact description of the library. For a raw library L , let F_L be its pointwise operating frontier: at each belief, it records the highest current expected payoff among retained strategies. Let C_L be its generative closure: the closed set of modules that can be assembled from retained carriers. Alongside the filtered belief, the pair

$$K_L = (F_L, C_L) \tag{1.1}$$

is the productive library state. The frontier records what can be operated now; the closure records what the modeled research process can use to produce later candidates. In the finite model, the raw generation, verification, admission, and library-update process projects onto this pair. Projects may have positive duration, and their terminal beliefs and admitted candidates may be correlated; neither feature requires the raw strategy identifiers once the frontier and closure are fixed.

This state reduction turns the organizing question into an exact source-relative compression problem. Give each active strategy a positive, fixed retention weight and write $W(L)$ for total burden. For a specified source library L , exact safe compression solves

$$\min_{L' \subseteq L} W(L') \quad \text{subject to} \quad F_{L'} = F_L, \quad C_{L'} = C_L, \quad s_0 \in L'. \tag{1.2}$$

The constraint is semantic rather than statistical: the retained sublibrary must reproduce both productive objects exactly. Because the feasible family is finite and contains the source, a minimum is attained. Every feasible sublibrary has the same modeled current operating rewards, research menus, candidate laws, future compressed updates, productive values, and available optimal actions as the source. Compression changes the raw representation and its burden, not the productive control process.

The exact constraint also gives a practical deletion certificate. Starting from the source, delete a proposed strategy only when recomputing the library shows that both its frontier and closure are unchanged. Rechecking after every accepted deletion produces a certified safe path: each intermediate library remains in the source's productive state. This certificate answers a local feasibility question, however, not the global minimization in (1.2). Earlier deletions can change which remaining strategies are redundant. An endpoint at which no single further deletion is safe can therefore have greater burden than another safe sublibrary reached by a different deletion order or found by complete search. Certified stepwise pruning and global minimum-resource compression are distinct decisions.

A simple bridge illustrates why the closure constraint is economically substantive. Suppose a strategy is dominated at every current belief but is the unique carrier of a module required by a

research project. If retained, that project can combine the module with other capabilities, survive and pass verification, and admit a valuable descendant after d periods. If the carrier is deleted, the present operating frontier is unchanged, but the project becomes unavailable and the descendant cannot be reached. When the discounted, survival-adjusted descendant payoff exceeds research cost, that difference is a positive delayed bridge margin. In the sharp construction, frontier-only pruning removes the complete attainable margin, so normalized loss equals one. The example does not say that every dominated strategy is valuable; it shows why current-payoff preservation alone cannot certify that a deletion is harmless.

Exact preservation is the primary problem. Two secondary variants ask what to retain when scarcity is allowed to change productive opportunities. Over a finite eligible catalog $\mathfrak{L}_\theta$, a hard capacity B selects the most productive library whose burden fits the constraint, while a resource price λ selects value net of burden:

$$\begin{aligned} V_\theta^*(b; B) &= \max_{L' \in \mathfrak{L}_\theta: W(L') \leq B} V_\theta(b, L'), \\ P_\theta(b; \lambda) &= \max_{L' \in \mathfrak{L}_\theta} \{V_\theta(b, L') - \lambda W(L')\}. \end{aligned} \tag{1.3}$$

These are outer, one-time retention choices, not repeated eviction decisions inside the Bellman problem. Their geometry is finite. Capacity value is attained, monotone, and constant between attainable burdens, but complementary modules can generate increasing unit increments. Penalized value is the continuous convex piecewise-affine upper envelope of finitely many value–burden lines, and selected burden falls weakly as the resource price rises. Optimal library composition can change discretely; no concavity, divisibility, or smooth policy path is imposed.

Productive value provides the link between exact compression and these resource variants. The value of inserting a strategy separates into the change in passive operating value and the change in the research-option premium. Along a named parameter path and within a fixed selected policy, the same accounting separates operational and generative sensitivity. Innovation duration summarizes the discount exposure of future invention value. These objects explain why a library is valuable without treating resource burden as itself an operating or generative payoff.

The results are organized in four groups.

1. *Raw-to-compressed state and exact safe compression.* The finite raw research process projects onto belief, operating frontier, and generative closure without changing productive values or available optimal actions. Within a source library, a minimum-resource sublibrary preserving the frontier and closure is attained (Theorems 3.2 and 4.1).
2. *Safe deletion, local versus global compression, and sharp loss.* Frontier–closure equality characterizes safe deletion under the stated detectability condition, and rechecking supplies a certified sequential path. A deletion-irreducible endpoint need not minimize global burden, while frontier-only pruning can attain the complete normalized delayed-bridge loss (Theorems 4.9 and 4.11).
3. *Finite resource optimization and productive-value interpretation.* Hard-capacity and resource-price selection have finite, potentially complementary and discrete geometry. Operational and generative value explain why the two safe-compression constraints matter, while named-path sensitivity and fixed-exposure timing remain auxiliary comparative statics (Theorems 4.13, 4.14 and 5.2 and Proposition C.1).
4. *Layered finite and financial evidence.* Exact fixtures and complete small-library enumeration check identities and boundary cases; registered randomized stress tests exercise the mechanisms

in frozen finite generators; and retrospective financial audits compare stepwise-safe, HiGHS-reported minimum-resource, and frontier-only libraries without using held-out descendant quality in a pruning rule or objective (Section 7).

The evidence levels are not interchangeable. Exact fixtures establish machine-rechecked facts about specified finite instances, not universal claims. Registered randomized results report frequencies for their frozen synthetic generators, not population estimates. Mixed-integer solver status is numerical global-search evidence and is reported separately from exact rechecks of selected libraries, frontiers, closures, and burdens. The financial exercises are retrospective information audits: structural and validation inputs are fixed before held-out descendant quality is read, and that quality is used only as an ex post opportunity diagnostic.

The scope is correspondingly narrow and explicit. States, beliefs, strategy catalogs, project menus, and retained libraries are finite. Retention weights are fixed and additive, and retention is chosen once outside the productive dynamic program. The generator and verifier factor through the modeled belief and frontier-closure state; provenance, age, multiplicity, time-varying maintenance burdens, endogenous eviction, and endogenous verification are not part of the primary model. Exact compression is source-relative and allows no frontier or closure tolerance. Sensitivities are reported only on named paths and, for elasticities, with positive denominators and a fixed selected policy; finite changes replace derivatives at discrete changes.

The remainder of the paper follows this dependency order. Section 2 fixes the boundary with partial-information control, policy compression, resource-constrained selection, maintained libraries, and generated financial strategies. Section 3 defines the raw process, the frontier-closure state, and retention burden. Section 4 develops exact global and certified local compression; Section 5 interprets productive value and finite capacity shadows; Section 6 solves the canonical finite control and resource problems; and Section 7 reports the controlled evidence. The primary appendices contain proofs, core sensitivities, computational verification, experiment-design information, and the verification-status summary. The Online Supplement contains extended comparative statics, complete numerical records, full randomized-study diagnostics, detailed financial-audit results, and reproducibility records.

2 Literature boundary

The paper draws on established ideas from partial-information and semi-Markov control, state and policy reduction, compact control libraries, generative action sets, evolving skill libraries, and automated financial-strategy construction. We treat each of those ingredients as prior art. The claimed boundary is narrower: their joint use in a finite model where a retained strategy can be both a current control and a carrier needed to generate later verified descendants. This section first locates the broad ingredients and then concentrates on the closest compression and library comparisons.

2.1 Control and representation

Belief-state reduction is standard in partially observable Markov decision processes. Finite-horizon POMDP values admit piecewise-linear convex representations, and incremental pruning removes redundant alpha vectors from fixed-model dynamic-programming backups [23, 2]. Bisimulation and related equivalence relations support state aggregation and model minimization when states agree on the behavior relevant to a specified comparison class [6]. Semi-Markov control likewise accommodates random decision durations. White develops a sufficient-statistic dynamic program for a finite-horizon, partially observed semi-Markov problem [27]; partial-information switching and regime-switching trading models use filtered beliefs to govern costly changes of operating mode [10, 4].

These results supply the control architecture used here. The paper’s frontier-closure statistic is a model-specific reduction of an augmented raw process, not a new general theory of POMDP sufficiency or semi-Markov timing. The additional issue is that the library is endogenous state: two libraries can have the same current payoff envelope but different parent-conditioned candidate laws. The operating frontier records current control value, while the generative closure records the modules available to the modeled research process. Standard alpha-vector pruning or MDP aggregation could be applied to a fully encoded augmented model; the present reduction identifies the smaller statistic implied by the paper’s finite generation, verification, and admission law.

Safety and statistical validation provide other preservation criteria. Finite-state-controller methods give high-probability policy improvement for POMDPs, while static policy libraries can support finite-horizon runtime-safety coverage [22, 9]. Reality-check and high-confidence policy-improvement methods address selection bias or the performance of evaluated policies [28, 24]. Those criteria remain important but distinct from exact equality of a source library’s current frontier and attainable descendant closure.

2.2 Compression and resource-limited libraries

Reward-free policy-space compression. Mutti, Del Col, and Restelli compress a parametric policy space into finitely many representatives whose induced state-action distributions provide reward-free Rényi-divergence coverage. Their formulation is related to set cover and is NP-hard in general [14]. This is the closest policy-compression comparison because it selects a compact policy set before downstream rewards are specified. Its preserved object is distributional representativeness. Here the feasible sets are sublibraries of a named source, and feasibility requires exact equality of two semantic objects. A currently dominated strategy can therefore be retained when it is the unique carrier of a module required by an attainable descendant. The two problems impose different covering relations even though both value compact policy sets.

Control-library optimization. Dey et al. select and order compact libraries of maneuvers or trajectories, using submodular-sequence structure to improve expected task performance or reduce search depth [5]. This is the closest classical control-library optimization problem: the library itself is the decision variable and computation is scarce. Its candidates are evaluated through static task usefulness. In the present model, retaining a candidate can also change the future candidate set, so module complementarities can matter even when current operating value does not. Consequently, exact safe compression is a finite combinatorial minimization under frontier-closure equality; rechecked single deletions certify feasibility but do not generally solve the global minimum-burden problem.

Metareasoning prices computations by their effect on later decisions, and adaptive-submodular formulations obtain greedy guarantees under diminishing stochastic marginal benefits [18, 8, 7]. These are the natural resource-allocation antecedents. The paper does not assume the diminishing returns needed for a generic greedy guarantee. Its hard-capacity and resource-price variants instead optimize over finitely many eligible libraries; complementarities and discrete changes in the selected library can produce nonconcave capacity increments and piecewise-affine penalized value.

Generative action sets. Xu et al. study online decision making when an agent can pay once to generate a new action that remains available, and derive a sublinear-regret algorithm for the creation-exploration-exploitation trade-off [30]. This establishes costly generation and persistent reuse as prior art. In that model, action generation expands the menu but is not conditioned on retaining particular existing actions as parents. Here that dependency changes the deletion problem: removal can leave every current action value unchanged while removing the modules needed to reach a later action. The distinction is parent-conditioned reachability, not the broad idea that actions can be generated.

2.3 Maintained and evolving skill libraries

Maintained skill libraries make parent and dependency structure operational. SkillFoundry builds validated executable skills from heterogeneous scientific resources and expands, repairs, merges, or prunes the resulting library [20]. SkillOps represents skills by typed contracts and organizes dependency, compatibility, validation, and maintenance signals in a library-level ecosystem graph [17]. Programmatic Skill Networks link executable programs through invocations and use duplicate removal, link redirection, branch pruning, and rollback validation during evolution [21]. These systems are the closest dependency-aware comparisons because removing one entry can affect later composition, retrieval, or construction rather than only present task performance.

Related systems recursively evolve skill banks, maintain typed skill graphs, construct hierarchical or versioned registries, compile external knowledge into reusable procedures, and synthesize code policies from validated components [29, 11, 25, 16, 3]. Together they establish maintained, evolving, compositional skill libraries as prior art. Their maintenance objectives are benchmark success, retrieval quality, validation, latency, compatibility, or lifecycle health. The present abstraction takes a fixed finite generation-verification-admission law as primitive and asks a different audit question: which source entries can be removed while exactly preserving both the operating frontier and every modeled feasible admitted descendant? It is not a replacement for engineering criteria concerning tests, provenance, repair, retrieval, or rollback.

2.4 Generated financial strategies

Automated financial-rule construction also has a long history. Genetic search has been used to generate technical trading rules, and direct reinforcement learning to construct trading policies [1, 13]. More recent systems add persistent libraries, reusable experience, and verification. FactorMiner combines formulaic factors with reusable evaluation skills and experience memory; ReCAP maintains regime-specific portfolio policies and combines them through a regime gate [26, 15]. These are substantive financial library precedents: one uses retained experience to guide later generation, while the other maintains a regime-specific operating policy library.

AlgoEvolve and EVOQUANT are the closest generated-strategy systems. AlgoEvolve evolves executable trading programs and uses selected programs as parents of later mutations; EVOQUANT applies controlled edits, provenance, verification gates, and retained optimization experience [19, 12]. They demonstrate that financial strategies can be generated, empirically screened, retained, and used in later search. The present paper begins after the finite generator and admission law have been specified and studies source-relative retention under partial information. Its retrospective financial audits are therefore compression diagnostics, not evidence of causal improvement, forecasting ability, market alpha, or deployable performance.

Joint claim boundary. This paper does not claim novelty for belief-state or semi-Markov control, policy or control-library compression, generated actions, evolving skill libraries, or automated financial-strategy generation. Its claim concerns the joint finite-model treatment. The raw partial-information process projects to belief, operating frontier, and generative closure; exact source-relative compression then minimizes retention burden while preserving both productive objects and the modeled control process. Rechecked frontier-closure equality certifies sequential deletion, although a deletion-irreducible endpoint need not be globally minimum-resource, and a sharp bridge construction shows that frontier-only pruning can lose the complete attainable delayed-descendant margin. Over finite eligible libraries, hard-capacity and resource-price selection yield attained but potentially nonconcave, complementary, and discrete geometry, while productive value separates operational and generative channels within fixed regimes. Exact fixtures, registered synthetic stress tests, and retrospective financial audits examine these mechanisms at distinct evidence levels. The claimed boundary is this package as a unit, not any ingredient in isolation.

3 Dynamic library model and sufficient state

The model is specified first on raw verified libraries. Research then generates and verifies a catalog candidate, admission updates the raw library, and the compressed process is obtained by projecting that raw evolution.

3.1 Hidden market state and filtered belief

Let $X = \text{Fin}(m)$, $2 \leq m$, be the hidden market-state type and let B be a nonempty finite grid of filtered beliefs. The controller observes $b \in B$ rather than $x \in X$. Each grid point has an injective rational interpretation $b \mapsto \mu_b$, at least one μ_b is non-Dirac, and the already-filtered belief evolves under an exact rational Markov kernel P_B . Economically, x is the unobserved market regime and b is the controller’s decision-relevant assessment of that regime. All carriers below are finite and have decidable equality; the probability-table and normalization conventions are in Appendix A.1.

3.2 Strategy operating profiles

Each strategy identifier $s \in S$ has an immutable state-contingent operating payoff $u_s : X \rightarrow \mathbb{Q}$. Its belief-indexed *operational profile* is

$$j_s(b) := \sum_{x \in X} \mu_b(x) u_s(x). \quad (3.1)$$

Thus $j_s(b)$ is the expected payoff from operating s now; project and continuation values are introduced separately below. A distinguished inactive strategy s_0 has zero operating payoff.

3.3 Modules and closure

Each strategy also carries an immutable finite set of modules, so its complete catalog row is

$$\sigma_s = (s, u_s, \text{mods}(s)), \quad \text{mods}(s) \in \mathcal{P}_{\text{fin}}(M), \quad \text{mods}(s_0) = \emptyset.$$

Economically, modules are reusable capabilities: they do not enter operating payoff directly, but they can be combined to support later research. The set of capabilities available after all such combinations is represented by an extensive, monotone, and idempotent closure operator

$$\text{cl} : \mathcal{P}_{\text{fin}}(M) \rightarrow \mathcal{P}_{\text{fin}}(M).$$

3.4 Raw library

A *raw verified library* is a set $L \subseteq S$ containing s_0 . Economically, L is the catalog of strategy identities currently retained and verified. Define

$$F_L(b) := \max_{s \in L} j_s(b), \quad U_L := \bigcup_{s \in L} \text{mods}(s), \quad (3.2)$$

$$C_L := \text{cl}(U_L). \quad (3.3)$$

Here F_L is the best currently attainable operating profile, U_L is the raw module inventory, and C_L is the resulting set of generative capabilities. The raw decision state is (b, L) . Continue earns $F_L(b)$, advances the belief once under P_B , and leaves L unchanged. The elementary frontier and closure calculus is recorded in Lemma A.1.

3.5 Positive-duration generation, verification, and admission

Economically, a research project uses the incumbent library's capabilities over a positive number of periods to seek a new verified strategy. Each project $q \in Q$ has prerequisites $\text{req}(q) \in \mathcal{P}_{\text{fin}}(M)$. At initiation belief b and closure C , candidate generation follows

$$G(q, b, C) \in \Delta_{\mathbb{Q}}(\text{Option}(S)),$$

with the point mass on none when the prerequisites are not contained in C . A generated s passes verification with probability $\nu(q, b, C, s)$. The admitted law is derived from these two stages:

$$\Gamma(q, b, C)(\text{some}(s)) = G(q, b, C)(\text{some}(s))\nu(q, b, C, s),$$

with all failed-generation and failed-verification mass assigned to none. Because G sees only (q, b, C) , generation cannot inspect raw identifiers, multiplicity, provenance, age, or admission order.

For a frontier–closure pair $K = (F, C)$, let $A(K)$ be the available finite project menu. A project $q \in A(K)$ has initiation cost $\kappa_q(b, K) \geq 0$, strictly positive duration d_q , and an operation flag $o_q \in \{0, 1\}$. Its exact joint completion law $\Lambda_q(\mathbf{b}, o \mid b, K)$ couples the P_B -belief path with the admitted outcome $o \sim \Gamma(q, b, C)$; the components may be correlated. During research the incumbent earns

$$-\kappa_q(b, K) + \sum_{t=0}^{d_q-1} \beta^t o_q F(b_t), \quad 0 \leq \beta < 1.$$

The old library remains in force through date $d_q - 1$, the outcome is applied at date d_q , and continuation starts from the terminal belief. Appendix A gives the full path marginals and normalization identities.

3.6 Raw update

Economically, successful admission adds one verified candidate, whereas every other outcome leaves the incumbent catalog unchanged. Formally,

$$L \oplus \text{none} = L, \quad L \oplus \text{some}(s) = L \cup \{s\}. \quad (3.4)$$

The corresponding raw terminal law is the pushforward

$$\mathcal{Q}_q^{\text{raw}}(\cdot \mid b, L) := (\mathbf{b}, o \mapsto (b_{d_q}, L \oplus o))_{\#} \Lambda_q(\cdot \mid b, (F_L, C_L)). \quad (3.5)$$

3.7 Compressed frontier–closure state

Definition 3.1 (Compressed frontier–closure state). The compressed innovation state is

$$K_L := (F_L, C_L). \quad (3.6)$$

Economically, the frontier records what the library can operate now and the closure records what it can help generate later. The realizable set $\mathcal{K} := \{K_L : L \text{ is a raw verified library}\}$ is finite because S is finite.

For an admitted outcome define

$$\text{addK}(K, \text{none}) = K, \quad \text{addK}((F, C), \text{some}(s)) = (b \mapsto \max\{F(b), j_s(b)\}, \text{cl}(C \cup \text{mods}(s))).$$

This map is the compressed effect of adding an admitted catalog row. Closure idempotence gives the local identity

$$K_{L \oplus o} = \text{addK}(K_L, o). \quad (3.7)$$

Applying $L \mapsto K_L$ to the raw law in Equation (3.5) yields the next-state laws $\bar{T}_q$ and $\bar{\mathcal{Q}}_q$: they are the pushforwards of Γ and Λ_q through addK and $(\mathbf{b}, o) \mapsto (b_{d_q}, \text{addK}(K, o))$, respectively. Thus the compressed transition is derived from the raw law, not separately postulated.

3.8 Projection theorem

Economically, two raw libraries with the same frontier–closure state support the same operating and research opportunities even when their retained strategy identities differ.

Theorem 3.2 (Raw-to-compressed controlled Markov projection). *In the finite exact model above:*

1. $\bar{T}_q$ and $\bar{\mathcal{Q}}_q$ are normalized and supported on realizable compressed states;

2. if $K_L = K_{\tilde{L}}$, every project has the same availability, cost, duration, operating-reward block, and joint terminal belief-next-compressed-state law from L and $\tilde{L}$;
3. at embedded decision epochs, (B_{τ_n}, K_{L_n}) is the controlled semi-Markov projection of (B_{τ_n}, L_n) ; and
4. for every finite calendar horizon h , belief b , and raw library L ,

$$V_h^{\text{raw}}(b, L) = \bar{V}_h(b, K_L).$$

The stationary raw and compressed Bellman fixed points also agree under $L \mapsto K_L$, and a maximizing stationary compressed selector lifts to a Bellman-optimal raw selector.

The projection, quotient, and restricted representation-refinement arguments are in Appendix A.2. Finiteness supplies normalization and attained action maxima, positive duration makes the calendar-horizon induction well founded, and contraction is used only for the stationary clauses. The proof leaves the joint completion law unfactorized, so it permits correlation between the belief path and admission outcome. Detectability does not enter this result. The Markov state is stated at decision epochs; during an unfinished project it must also record the project and remaining duration.

3.9 Resource burden and exact compression

Resource burden is an outer retention decision, not operating payoff or project cost. Each active strategy has a fixed rational weight $w_s > 0$, while $w_{s_0} = 0$, and the library's one-time retention burden is

$$W(L) := \sum_{s \in L} w_s. \quad (3.8)$$

Economically, $W(L)$ can commensurate storage, validation, maintenance, monitoring, retrieval, and governance requirements. If S_θ^{elig} is the finite outer-certified catalog, write $\mathfrak{L}_\theta = \{L' \subseteq S_\theta^{\text{elig}} : s_0 \in L'\}$. Later capacity and price problems optimize over this family. Equal compressed states can have unequal burdens, so a resource optimizer must retain the raw library L or augment K_L by $W(L)$.

For compression of a specified source L , define

$$\mathcal{P}_{\text{safe}}(L) := \{L' \subseteq L : s_0 \in L', K_{L'} = K_L\}. \quad (3.9)$$

This family contains exactly the source sublibraries that preserve both the current operating frontier and the generative closure. By the projection theorem, every member therefore has the same project menus, positive-duration research laws, future compressed updates, and productive values as the source. Resource minimization changes only the retained raw representation of that control process. The detectability result in Section 4 gives the corresponding observable converse. Exact safe compression is

$$\begin{array}{ll} \text{minimize} & W(L') \\ \text{subject to} & s_0 \in L', \\ & K_{L'} = K_L. \end{array} \quad (3.10)$$

4 Optimal Innovation-Safe Compression

Starting from a verified source library L , compression asks how little resource burden can be retained without changing what the controller can operate now or help generate later. Section 3 showed that

these productive requirements are summarized by $K_L = (F_L, C_L)$. Throughout this section, *safe* means source-relative preservation of that exact pair, *global minimum* means minimum burden over every safe sublibrary of the source, and an *optimizer* in the capacity or penalized problems means a maximizer over the full outer-certified family $\mathfrak{L}_\theta$. The sequence moves from exact representation and certified deletion to the local–global boundary, the frontier-only loss, and the two outer selection problems.

4.1 Minimum-resource safe representation

Economic question. Among all retained sublibraries that preserve both current operating capability and future generative capability, what is the minimum retention burden and which representations attain it?

Recall the source-relative family

$$\mathcal{P}_{\text{safe}}(L) = \{L' \subseteq L : s_0 \in L', K_{L'} = K_L\}.$$

It is a projection fiber restricted to sublibraries of L . Its global minimum-weight correspondence is

$$\mathcal{M}_{\text{safe}}(L) := \arg \min_{L' \in \mathcal{P}_{\text{safe}}(L)} W(L'). \quad (4.1)$$

Theorem 4.1 (Minimum-resource innovation-safe representation). *For every finite source library, $\mathcal{P}_{\text{safe}}(L)$ is nonempty and finite, so $\mathcal{M}_{\text{safe}}(L) \neq \emptyset$. Every two safe-feasible libraries have the same compressed state and hence the same finite-horizon productive values. Under the discounted contraction, they also have the same stationary productive value and the same complete set of optimal stationary actions.*

The theorem proves existence and productive preservation. The source itself makes the feasible family nonempty, finiteness makes the minimum attained, and $K_{L'} = K_L$ carries every feasible representation to the same productive process. Contraction is needed only for stationary value and action preservation; detectability and probabilistic independence are not used. This is not a uniqueness or pruning theorem: it neither singles out one minimum-weight representation nor says that a deletion rule reaches one. Appendix B gives the finite argument.

For identity closure, exact safe feasibility has a useful binary cover form. Let $x_s \in \{0, 1\}$ indicate retention, fix $x_{s_0} = 1$, and define the source attainers and module carriers by

$$A_b := \{s \in L : j_s(b) = F_L(b)\}, \quad R_m := \{s \in L : m \in \text{mods}(s)\}.$$

Safe feasibility is then equivalent to

$$\sum_{s \in A_b} x_s \geq 1 \quad (b \in B), \quad \sum_{s \in R_m} x_s \geq 1 \quad (m \in C_L), \quad (4.2)$$

with objective $\min \sum_s w_s x_s$. For general closure, the second family of rows is replaced by the exact equality $\text{cl}(\bigcup_{s: x_s=1} \text{mods}(s)) = C_L$; independent carrier rows would miss closure complementarities.

4.2 Innovation-safe deletion certificate

Economic question. When can one retained strategy be removed without changing either current operating opportunities or future research opportunities?

Definition 4.2 (Deletion redundancy). For $s \in L \setminus \{s_0\}$, write $L^{-s} = L \setminus \{s\}$. Deletion has two distinct certificates:

$$F_{L^{-s}} = F_L \quad (\text{operational redundancy}), \quad (4.3)$$

$$C_{L^{-s}} = C_L \quad (\text{generative redundancy}). \quad (4.4)$$

The first compares the entire belief-indexed frontier, not only the current payoff. The second compares closed capabilities, not only duplicated raw module rows.

To state the observable converse, let $L \sim_{\text{DI}} L'$ mean that the two libraries have the same frontier and, for every belief and project, the same availability-tagged cost, duration, joint terminal-belief/next-compressed-state law, and incumbent operating-reward block.

Definition 4.3 (Dynamic innovation equivalence). With $\langle z \rangle_{A(K)} = \text{some}(z)$ when $q \in A(K)$ and none otherwise, $L \sim_{\text{DI}} L'$ means equality of the five observations

$$\left(F_L(b), \langle \kappa_q(b, K_L) \rangle_{A(K_L)}, \langle d_q \rangle_{A(K_L)}, \langle \overline{\mathcal{Q}}_q(b, K_L) \rangle_{A(K_L)}, \langle R_q^{\text{op}}(b, K_L) \rangle_{A(K_L)} \right) \quad (4.5)$$

with their L' counterparts for every b, q .

Assumption 4.4 (Observable raw closure detectability). If two realizable states have the same frontier but different closures, some project changes an availability-tagged cost, duration, or joint terminal law after generation, admission, raw update, and projection.

Detectability is used only for necessity. Without it, a module may be present but behaviorally silent in every available experiment.

Theorem 4.5 (Frontier-closure characterization and safe deletion). *Under the raw factorization of Section 3,*

$$K_L = K_{L'} \implies L \sim_{\text{DI}} L'.$$

Under Assumption 4.4, the implication is an equivalence. Consequently, for every $s \in L \setminus \{s_0\}$,

$$K_{L^{-s}} = K_L \iff \begin{cases} F_{L^{-s}} = F_L, \\ C_{L^{-s}} = C_L. \end{cases} \quad (4.6)$$

The two equalities certify innovation-safe deletion and preserve every finite-horizon productive value. Under detectability they are also necessary for dynamic innovation equivalence; under the discounted contraction they preserve stationary value, all stationary action comparisons, and the full optimal-action correspondence.

The theorem proves that the two deletion tests are exactly compressed-state preservation. Under the Section 3 factorization, they are sufficient for dynamic innovation equivalence and productive-value preservation; under detectability, they are also necessary for the observable equivalence. The certificate is local to one candidate in the current library. If generation could inspect hidden raw identifiers, sufficiency would fail; without detectability, only the necessity direction would fail. Neither direction factors the joint completion law, and contraction enters only after state equality to obtain the stationary clauses. The bridge in Theorem 4.11 shows where the operational-only test fails: operational redundancy alone does not imply generative redundancy. Appendix A records the assumption-boundary examples and quotient refinement.

4.3 Rechecked pruning

Economic question. How can repeated deletions remain certified when removing one strategy may make another strategy essential?

The deletion certificate yields the following reduction routine.

Rechecked innovation-safe pruning. Start at $L_0 = L$. At step i , choose an active $s \in L_i$, recompute $F_{L_i - s}$ and $C_{L_i - s}$, and erase s only if both equal their current values. Restart the scan after every accepted deletion and stop only after no retained active strategy passes both tests. Return the endpoint and the complete deletion trace.

Corollary 4.6 (Rechecked pruning certificate). *For every rechecked trace $L_0 \supseteq L_1 \supseteq \dots \supseteq L_m$,*

$$K_{L_m} = K_{L_0}, \quad L_m \sim_{\text{DI}} L_0.$$

Every accepted deletion is safe, the endpoint remains in $\mathcal{P}_{\text{safe}}(L_0)$, and all productive values are preserved. If the final scan is complete, its failure to find another safe erasure is a certificate of one-deletion irreducibility.

The corollary proves feasibility of the entire path by composing the rechecked compressed-state equalities. A valid trace alone does not prove that the endpoint is irreducible; that conclusion additionally requires the complete final scan specified above.

Call a safe library E *one-deletion irreducible* when no retained active s satisfies $K_{E-s} = K_L$. Call E *inclusion-wise irreducible* when no proper sublibrary of E belongs to $\mathcal{P}_{\text{safe}}(L)$. In this monotone exact-safe fiber the two notions coincide: any proper safe sublibrary implies a safe one-element deletion by frontier and closure monotonicity. A complete rechecked scan therefore certifies both notions. This remains a local certificate: neither notion implies global minimum weight.

Proposition 4.7 (Certified pruning specification). *A pruning output is certified only together with its rechecked trace; the trace certifies each deletion and endpoint productive-value preservation.*

Example 4.8 (Why deletion certificates must be rechecked). If two strategies are the only carriers of a module, each may be redundant initially, but after deleting one the survivor is essential. Applying both stale certificates would remove the module from closure.

4.4 Local versus global optimality

Economic question. Does the absence of another certified one-strategy deletion imply that the retained library has minimum burden?

A library is globally minimum weight precisely when it belongs to $\mathcal{M}_{\text{safe}}(L)$ in (4.1); it must be compared with every safe sublibrary, not only its one-deletion neighbors.

Theorem 4.9 (Local/global boundary). *With strictly positive active weights, every global minimum is one-deletion irreducible. Every rechecked deletion trace is safe, but a one-deletion- irreducible endpoint need not be globally minimum weight. Even deleting the unique heaviest currently safe strategy can lead to a strictly heavier endpoint than another safe representation.*

The theorem proves only the forward local implication: a global minimum-weight safe representation cannot contain a safely deletable positive-weight strategy. Its converse is false, and the following exact (2, 2, 3) construction shows that even a uniquely determined heaviest-safe-first step can terminate above the global minimum.

Example 4.10 (Worked example: certified greedy endpoint versus global minimum). Take one belief, identity closure, and three zero-profile active strategies. Strategies s_1 and s_2 carry $\{m_1\}$ and $\{m_2\}$ with weights 2 and 2; the bundle s_{12} carries $\{m_1, m_2\}$ with weight 3. The inactive strategy has zero profile, no modules, and zero weight. The current library is $L = \{s_0, s_1, s_2, s_{12}\}$.

Representation	Active strategies	Burden (units)	Frontier (payoff units)	Closure
Current library	s_1, s_2, s_{12}	7	$F \equiv 0$	$\{m_1, m_2\}$
Heaviest-safe-first endpoint	s_1, s_2	4	$F \equiv 0$	$\{m_1, m_2\}$
Minimum-weight safe library	s_{12}	3	$F \equiv 0$	$\{m_1, m_2\}$

All displayed values are exact rationals; denominator-one values are rendered as integers.

At the source, s_{12} is the unique heaviest safely deletable strategy. Deleting it first saves 3 units and leaves the inclusion-wise irreducible pair $\{s_1, s_2\}$; neither singleton can then be removed. Complete enumeration of the five safe sublibraries shows that $\{s_0, s_{12}\}$ is the minimum-weight safe representation. It saves 4 units relative to the current library, one more than the greedy endpoint. Both endpoints have exactly the source frontier and closure, so their productive values agree; only resource burden differs. Deleting the two singletons before the bundle reaches the global solution, making the order dependence explicit.

The registered counterexample establishes the clean separation

$$\text{certified one-deletion irreducibility} \not\Rightarrow \text{global minimum-weight compression.}$$

It does not provide a greedy approximation ratio or imply deletion-order invariance. Appendix B gives the exhaustive five-library comparison and the general minimum-attainment and global-to-local arguments. Those arguments use finiteness for attainment and positive active weights for the global-to-local implication; they use neither detectability nor probabilistic independence.

4.5 Sharp frontier-only relaxation loss

Economic question. What productive value can be lost if compression preserves the current operating frontier but ignores the capabilities needed to generate descendants?

Frontier-only pruning replaces the exact constraint $K_{L'} = K_L$ by $F_{L'} = F_L$. The relaxation is operationally exact today but may remove a unique module carrier and therefore a positive-mass path to a valuable descendant. The bridge construction isolates that loss.

Let $L = \{s_0, s_L, s_R, s_M\}$ have frontier $F_L = (3, 3)$ on two beliefs and closure $C_L = \{m\}$. The profiles of s_L and s_R attain the two frontier coordinates, while the dominated bridge s_M has profile $(1, 1)$ and uniquely carries m . Hence s_M is operationally redundant but generatively essential. Deleting it preserves F_L , removes m , and disables the only project leading to descendant c .

For this canonical one-project comparison, define Loss_d as the retained bridge envelope minus its frontier-only comparator. The reward cap C bounds the complete distinguished terminal continuation gain, there are no other differential projects or continuation gains, and the incumbent operating block is common.

Theorem 4.11 (Sharp normalized frontier-only relaxation loss). *Fix exact rationals $0 \leq \beta < 1$, $0 \leq \rho, \pi \leq 1$, reward cap $C \geq 0$, cost $\kappa \geq 0$, and delay $d \geq 1$. Suppose the dominated bridge uniquely enables a descendant, retention gives survival mass ρ^d and admission probability π , and deletion makes that descendant mass zero. If $\kappa \leq \beta^d \rho^d \pi C$, then*

$$\text{Loss}_d = \beta^d \rho^d \pi C - \kappa. \quad (4.7)$$

For fixed $(\beta, \rho, \pi, d, \kappa)$, this is the sharp upper bound over descendant rewards in $[0, C]$, attained at C . Whenever the denominator is positive,

$$\frac{\text{Loss}_d}{\beta^d \rho^d \pi C - \kappa} = 1.$$

Thus the relaxation can destroy the entire attainable net bridge value even though it leaves the current frontier unchanged. If $C \leq 1$, the additive loss is at most one; arbitrarily large loss requires reward scaling.

The theorem proves an exact and sharp loss statement for the displayed canonical one-project comparison. The bridge is operationally redundant because deleting it preserves F_L , but generatively essential because its deletion removes m from C_L and disables the descendant. The normalized loss is one whenever the attainable net bridge value is positive. This does not assert an unbounded normalized loss: under $C \leq 1$ the additive loss is at most one, and arbitrary additive loss requires the reward scaling in the corollary below.

Common incumbent operating blocks cancel under frontier equality. If operation differs during research, its exact discounted difference must be added. The factor $\rho^d \pi$ comes from the sequential raw generation and conditional admission rows, not from path–outcome independence. Finiteness is used only for the attained one-project maximum; detectability and contraction are not used. The catalog and proof are in Appendix B.

Corollary 4.12 (Arbitrary additive loss only by scaling). *For $M \in \mathbb{Q}_{\geq 0}$, take $(d, \beta, \rho, \pi, \kappa, C) = (1, 1/2, 1, 1, 0, 2M)$. Then $\text{Loss}_1 = M$.*

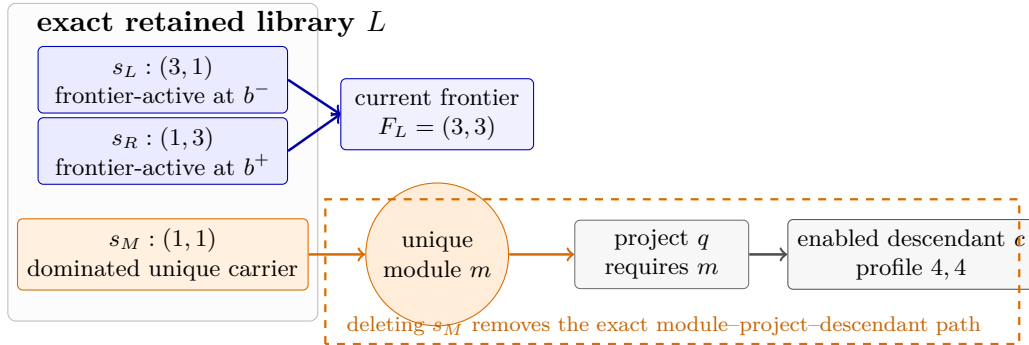


Figure 1: Exact bridge-mechanism fixture, not a statistical estimate. The frontier-only relaxation deletes the dominated s_M while preserving the two-belief frontier $F_L = (3, 3)$, but removes its unique module m , disables project q , and destroys access to the profile-(4,4) descendant. Profiles and dependencies are exact source rows; diagram coordinates are presentation-only and imply no additional path.

4.6 Capacity-constrained selection

Economic question. If available retention capacity is too small to preserve a specified source library's productive state, what productive value is attainable and which outer-certified libraries attain it?

Exact compression starts from a specified source and preserves its productive state. Capacity selection instead ranges over the entire finite outer-certified family $\mathcal{L}_\theta$ and permits productive tradeoffs:

$$V_\theta^*(b; B) := \max_{L' \in \mathcal{L}_\theta: W(L') \leq B} V_\theta(b, L'). \quad (4.8)$$

The budget is imposed once at retention; it does not alter the raw Bellman recursion or later admission update.

Theorem 4.13 (Finite capacity-constrained retention). *Suppose $\mathfrak{L}_\theta$ contains the zero-burden inactive-only library. For every $B \geq 0$, the maximum in (4.8) is attained. If $B_1 \leq B_2$, then*

$$V_\theta^*(b; B_1) \leq V_\theta^*(b; B_2).$$

The capacity value is constant between consecutive attainable burdens and can change only at an attainable burden; every exact forward capacity increment is nonnegative. Unit capacity increments need not diminish: an exact two-module complementarity instance has successive increments 0 and 1.

The theorem proves attainment, monotonicity, and finite step geometry for the capacity-maximized productive value. It does not prove uniqueness or inclusion nesting of capacity maximizers, and it does not imply diminishing capacity increments: the exact $(0, 0, 1)$ complementarity profile is the counterexample. Discreteness is substantive; an attainable hard-capacity maximizer may be unsupported by every resource price.

Throughout this theorem and its Lean formalization, the inactive-only library has burden zero. Section 6's canonical plots use a separately marked display translation $\widetilde{W} = 1 + W$; that convention does not change this theorem or its assumptions.

4.7 Penalized selection

Economic question. How does a scalar price on retention burden change optimized value and the burdens of maximizing outer-certified libraries?

Pricing burden gives the second outer problem

$$J_\theta^*(b; \lambda) := \max_{L' \in \mathfrak{L}_\theta} \{V_\theta(b, L') - \lambda W(L')\}, \quad \lambda \geq 0. \quad (4.9)$$

Each library supplies an affine branch $J_{L'}(\lambda) = V_\theta(b, L') - \lambda W(L')$.

Theorem 4.14 (Finite penalized value envelope). *For a fixed nonempty finite eligible family, the maximum in (4.9) is attained. The optimized value is continuous, nonincreasing, convex, and piecewise affine in the real price λ , with switching candidates among finitely many pairwise branch intersections. If $0 \leq \lambda_1 < \lambda_2$, every optimizer at λ_2 has weakly lower burden than every optimizer at λ_1 . Wherever one branch is strictly dominant, the local slope is $-W(L')$.*

The theorem proves the geometry of the optimized penalized-value envelope and the cross-price order of optimizer burdens. It orders burdens, not raw library identities: a higher-price optimizer need not be a sublibrary of a lower-price optimizer.

For two unequal-burden libraries, their candidate crossing price is

$$\lambda_{12} = \frac{V_\theta(b, L_1) - V_\theta(b, L_2)}{W(L_1) - W(L_2)}. \quad (4.10)$$

It is an actual envelope breakpoint only if the two branches both maximize the penalized objective over the full eligible family there. Equal-burden branches are parallel or coincident. At a tie, the optimizer is set-valued; the theorem does not assert differentiability at a breakpoint. Appendix B gives the affine-envelope and burden-order arguments. Online Supplement S3 records the optional conditional replacement formulation outside the theorem sequence above.

5 Value Interpretation

Section 4 determines which strategies remain. This section explains why its two preservation objects have economic value. It keeps the value decomposition, one descendant-event bound, and the finite shadow value needed to interpret the compression problem. Appendix C contains the core sensitivity formulas; extended belief-occupation and interaction results are in Online Supplements S1–S3. Resource burden is not a third productive channel: it enters through the outer capacity or penalized problem of Section 4.

5.1 Operational–generative decomposition

Use the unified raw process of Theorem 3.2. Its *passive value* freezes the library and rules out research:

$$\begin{aligned} P_0(b, L) &= 0, \\ P_{n+1}(b, L) &= F_L(b) + \beta \sum_{b'} P_B(b, b') P_n(b', L). \end{aligned} \quad (5.1)$$

Let $U_n(b, L)$ be the corresponding full value with Continue and every feasible positive-duration research project, and let $\Omega_n(b, L) := U_n(b, L) - P_n(b, L)$ be the research-option premium.

Definition 5.1 (Insertion-value channels). For a candidate s , define

$$\mathcal{I}_n(s \mid b, L) := U_n(b, L \cup \{s\}) - U_n(b, L), \quad (5.2)$$

$$\Delta_n^{\text{op}}(s \mid b, L) := P_n(b, L \cup \{s\}) - P_n(b, L), \quad (5.3)$$

$$\Delta_n^{\text{gen}}(s \mid b, L) := \Omega_n(b, L \cup \{s\}) - \Omega_n(b, L). \quad (5.4)$$

Theorem 5.2 (Operational–generative value decomposition). *For every finite horizon, belief, raw library, and candidate,*

$$\mathcal{I}_n(s \mid b, L) = \Delta_n^{\text{op}}(s \mid b, L) + \Delta_n^{\text{gen}}(s \mid b, L). \quad (5.5)$$

This is an accounting identity, not a sign claim. The operational term asks what insertion changes when invention is switched off. The generative term is the residual change in the option to research, so it can be negative if insertion changes costs, completion laws, or the relative appeal of projects. The projection theorem gives $U_n(b, L) = V_n(b, K_L)$, allowing both channels to be evaluated on the sufficient frontier–closure state.

Corollary 5.3 (Frontier-silent insertion). *If $F_{L \cup \{s\}} = F_L$, then $\Delta_n^{\text{op}}(s \mid b, L) = 0$. If also $C_{L \cup \{s\}} = C_L$, then $\mathcal{I}_n(s \mid b, L) = 0$.*

Example 5.4 (Operationally silent, generatively valuable bridge). A zero-profile bridge carries the unique module enabling a duration-one, zero-cost project whose descendant pays two. With one belief, suspended operation, and $\beta = 1/2$, insertion gives

$$\Delta_2^{\text{op}} = 0, \quad \Delta_2^{\text{gen}} = 1. \quad (5.6)$$

The strategy is silent on the current frontier and valuable solely as an input into future invention.

5.2 Joint descendant-event lower bound

The generative channel becomes economically sharp when one retained carrier opens a specific path to a descendant. Write $L^+ = L \cup \{s\}$. For a project q of duration $d \leq h$ and a distinguished descendant g , define the terminal-belief/descendant event mass from the unified joint law:

$$\eta_{q,g}(b' \mid b, K) := \Pr(B_d = b', O = \text{some}(g) \mid B_0 = b, K, q). \quad (5.7)$$

This is a subprobability mass and permits arbitrary correlation between the terminal belief and admission. The operating-timing adjustment is

$$A_{q,h}^{\text{op}}(b, L^+) := \mathbb{E}_{b, K_{L^+}}^q \left[G^{\text{op}}(K_{L^+}, q, \mathbf{B}) + \beta^d P_{h-d}(B_d, L^+) \right] - P_h(b, L^+). \quad (5.8)$$

It records operating value earned or forgone while research consumes time.

Theorem 5.5 (Joint descendant-event generative-option lower bound). *Suppose $F_{L^+} = F_L$; q , of duration $d \leq h$, is feasible at L^+ and infeasible at L ; its cost is κ ; and the deleted comparator has $\Omega_h(b, L) = 0$. Let $G(b') \geq 0$ lower-bound the complete full-minus-passive continuation gain after admitting g on every length- d path. If failure leaves the library unchanged, other admissions are insertion-only, and full continuation retains the passive policy, then*

$$\Delta_h^{\text{gen}}(s \mid b, L) \geq \max \left\{ -\kappa + A_{q,h}^{\text{op}}(b, L^+) + \beta^d \sum_{b'} \eta_{q,g}(b' \mid b, K_{L^+}) G(b'), 0 \right\}. \quad (5.9)$$

If $A_{q,h}^{\text{op}} = 0$, this reduces to

$$\Delta_h^{\text{gen}}(s \mid b, L) \geq \max \left\{ -\kappa + \beta^d \sum_{b'} \eta_{q,g}(b' \mid b, K_{L^+}) G(b'), 0 \right\}. \quad (5.10)$$

Under the event-specific factorization

$$\eta_{q,g}(\cdot \mid b, K_{L^+}) = \pi \rho^d \mu_{q,d}(b, \cdot),$$

write $\overline{G}_{q,d}(b) := \sum_{b'} \mu_{q,d}(b, b') G(b')$.

Corollary 5.6 (Product specialization and signs). *Under that factorization, the terminal term in (5.9) is $\beta^d \pi \rho^d \overline{G}_{q,d}(b)$. Holding other terms fixed, the guarantee rises with pointwise joint mass, pointwise gain, and the operating adjustment and falls with cost. Total descendant mass alone has no sign if its terminal-belief distribution also changes.*

The joint bound does not require independence. Duration has no unconditional sign because it can change discounting, the joint law, belief occupation, operating rewards, continuation gains, and horizon feasibility. Appendix A gives the proof and exact retained-carrier example; Online Supplement S3 gives the detailed delay algebra.

5.3 Capacity shadow value

The hard-capacity problem in Equation (4.8) yields the exact forward shadow value

$$\Delta_B^\delta V_\theta^*(b; B) := V_\theta^*(b; B + \delta) - V_\theta^*(b; B), \quad \delta > 0. \quad (5.11)$$

By Theorem 4.13, this finite difference is nonnegative and is zero whenever the interval crosses no value-changing attainable burden. It is the appropriate marginal object because capacity value is a finite step function, not a differentiable relaxation.

Scarce capabilities can be complementary. In the exact two-unit witness, neither first capability is productive alone but the pair unlocks value one, so the unit shadow sequence is $0, 1$. Capacity value therefore need not be concave and marginal capacity values need not diminish. Conversely, additive independent gains with equal unit weights give weakly decreasing grid shadows. Any normalized capacity arc additionally requires positive base capacity and positive base value; when either denominator is zero, we report (5.11) itself.

5.4 Bridge-margin sensitivity

The bridge comparison also gives one useful sensitivity warning. Its gross descendant value is $A_{\text{br}} = \beta^d \rho^d \pi C$, its signed net margin is $M_{\text{br}} = A_{\text{br}} - \kappa$, and its normalized positive margin is $m_{\text{br}} = M_{\text{br}}/A_{\text{br}}$. On the positive-value branch, named-coordinate elasticities are amplified by $1/m_{\text{br}}$, so the bridge becomes locally fragile as its normalized margin approaches zero. Vanishing value scale alone is not enough, and at or below the threshold ordinary percentage elasticity is undefined. Appendix C gives the core sensitivity formulas and switching conventions; extended belief-occupation, coverage, persistence, and frontier-closure interaction results are in Online Supplements S1–S3.

6 Dynamic solution and canonical optimization model

The resource problems in Sections 4–5 take productive library values as inputs. This section closes the model by solving those values from the same raw admission and update laws used to define the sufficient state. It then places exact safe compression, hard capacity, resource pricing, and elasticity on one six-state rational benchmark. Exact arithmetic comes first throughout; `Float64` is used only for the final numerical diagnostic.

6.1 Unified Bellman operator

Write $K = K_L = (F_L, C_L)$, let $A(K)$ be the finite feasible project menu, and let $d(q) > 0$ be the duration of project q . Conditional on $B_0 = b$, the belief-path marginal during research is

$$\Pr(B_{0:d(q)} = b_{0:d(q)} \mid b) = \mathbf{1}\{b_0 = b\} \prod_{t=0}^{d(q)-1} P_B(b_t, b_{t+1}). \quad (6.1)$$

Let $\Lambda_q(\cdot \mid b, K)$ be the exact joint law of this path and the admitted outcome O ; path and admission need not be independent. During research, incumbent operation contributes

$$\mathcal{O}_q(B_{0:d(q)-1}; K) := \sum_{t=0}^{d(q)-1} \beta^t \mathbf{1}\{o(q) = 1\} F_K(B_t). \quad (6.2)$$

Definition 6.1 (Unified Continue, research, and Bellman operators). For $V : B \times \mathcal{K}_{\text{real}} \rightarrow \mathbb{R}$, define

$$(\text{CV})(b, K) := F_K(b) + \beta \sum_{b'} P_B(b, b') V(b', K), \quad (6.3)$$

$$(\text{R}_q V)(b, K) := -\kappa(b, K, q) + \mathbb{E}_{\Lambda_q(\cdot \mid b, K)} \left[\mathcal{O}_q(B_{0:d(q)-1}; K) + \beta^{d(q)} V(B_{d(q)}, \text{addK}(K, O)) \right], \quad (6.4)$$

$$(\text{TV})(b, K) := \max \left\{ (\text{CV})(b, K), \max_{q \in A(K)} (\text{R}_q V)(b, K) \right\}. \quad (6.5)$$

If $A(K) = \emptyset$, the inner research maximum is omitted, so $(\text{TV})(b, K) = (\text{CV})(b, K)$.

Continue consumes one date; research consumes $d(q)$ dates. Suspending operation removes the block in (6.2), not belief evolution or the terminal raw update. At a finite horizon only projects whose duration fits the remaining dates are feasible. The stationary equation is $V_\infty = \text{TV}_\infty$ on the complete menu.

Proposition 6.2 (Finite-state contraction and selectors). *Suppose all carriers are finite, primitive probabilities and payoffs are exact rationals, project durations are positive, and $0 \leq \beta < 1$. Every finite-horizon Bellman maximum is attained. The raw and compressed stationary operators are monotone sup-norm contractions with modulus β , have unique fixed points, and agree under raw-to-compressed projection. Hence dynamically innovation-equivalent libraries have equal finite- and infinite-horizon values. A stationary maximizing selector exists, satisfies $\text{T}_{\pi^*} V_\infty = V_\infty = V^{\pi^*}$, and lifts to a raw stationary selector. For any initial table v_0 ,*

$$\|\text{T}^n v_0 - V_\infty\|_\infty \leq \frac{\|v_0 - \text{T} v_0\|_\infty \beta^n}{1 - \beta}. \quad (6.6)$$

The contraction, intertwining, and selector argument is in Appendix A.2. Appendix D gives the numerical stopping rule and compact verification certificate; Online Supplement S4 retains the complete convergence history.

6.2 Unified six-state compressed benchmark

The benchmark is constructed from raw primitives. Its catalog is

s	$(j_s(\ell), j_s(h))$	$\text{mods}(s)$	role
s_0	$(0, 0)$	$\emptyset$	mandatory inactive policy
c_A	$(0, 0)$	$\{m\}$	silent carrier A
c_B	$(0, 0)$	$\{m\}$	silent carrier B
g	$(2, 4)$	$\{m\}$	profitable descendant and carrier.

Closure is the identity on $\{m\}$, and every raw library contains s_0 . Exhausting the eight subsets of $\{c_A, c_B, g\}$ produces three compressed library states,

$$K_0 = ((0, 0), \emptyset), \quad K_1 = ((0, 0), \{m\}), \quad K_2 = ((2, 4), \{m\}).$$

Together with $B = \{\ell, h\}$, these are the six Bellman states. The exact belief law and discount are

$$P_B = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \quad P_B^2 = \begin{pmatrix} 5/8 & 3/8 \\ 3/8 & 5/8 \end{pmatrix}, \quad \beta = 1/2. \quad (6.7)$$

Discover is available at K_0 , lasts one date, operates the incumbent, and costs $1/32$ at ℓ and 1 at h . Raw generation and verification give $\Gamma_D = \frac{1}{4}\delta_\perp + \frac{1}{2}\delta_{c_A} + \frac{1}{4}\delta_{c_B}$. Scale is available at K_1, K_2 , lasts two dates, operates the incumbent, costs $3/16$, and admits g with marginal probability $3/4$. Its success is correlated with the terminal belief: the exact conditional probability is $3/5$ for $(B_0, B_2) = (\ell, \ell)$, $1/3$ for (h, ℓ) , and 1 when $B_2 = h$. Applying the raw update $L \oplus s = L \cup \{s\}$ and only then projecting gives

$$R_D(K_0) = \frac{1}{4}\delta_{K_0} + \frac{3}{4}\delta_{K_1}, \quad R_S(K_1) = \frac{1}{4}\delta_{K_1} + \frac{3}{4}\delta_{K_2}, \quad R_S(K_2) = \delta_{K_2}.$$

Thus no transition is stipulated directly on the compressed carrier. The full path-by-admission atoms and reward-block algebra are in Online Supplement S4.

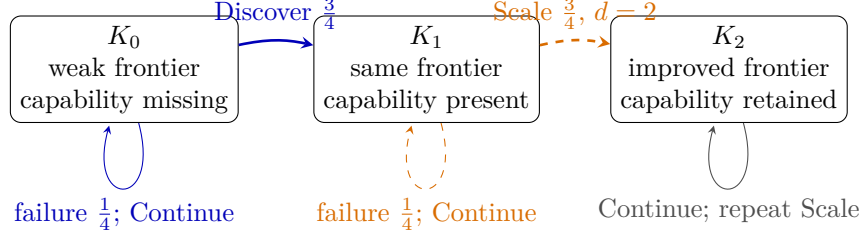


Figure 2: Exact raw-derived compressed transition law. Each edge label is the one-project probability obtained by applying raw generation, verification, admission, and library update before projection. Solid blue is Discover, dashed orange is Scale, and self-loops collect failure and Continue; no intermediate states or smooth interpolation are implied.

6.3 Exact rational dynamic solution

Exact raw and compressed policy iteration terminate after three improvement steps with policy-equation and Bellman residuals equal to zero. Projection matches values and actions on all eight raw libraries. The minimum exact winning margin is $3/16$, so the benchmark has no action tie.

State	Belief	V^* (payoff units)	π^*	Winning margin (payoff units)
K_0	ℓ	$\frac{115}{288}$	Discover	$\frac{23}{96}$
K_0	h	$\frac{23}{288}$	Continue	$\frac{245}{384}$
K_1	ℓ	$\frac{1}{288}$	Scale	$\frac{48}{23}$
K_1	h	$\frac{7}{6}$	Scale	$\frac{48}{29}$
K_2	ℓ	$\frac{14}{3}$	Continue	$\frac{3}{16}$
K_2	h	$\frac{22}{3}$	Continue	$\frac{3}{16}$

Table 1: Exact rational stationary solution of the raw-derived six-state benchmark. Values and winning margins are measured in the benchmark’s payoff units. Denominator-one rationals are rendered as integers. Appendix D reports the zero-residual certificate; Online Supplement S4 reports the full action-value record.

6.4 Resource weights and exact safe-compression solutions

Write I, A, B, D for s_0, c_A, c_B, g . The three prospectively fixed integer schedules keep the silent carriers symmetric. In the coordinate order (I, A, B, D) , their primary-model weight vectors are equal-active $(0, 1, 1, 1)$, carrier-heavy $(0, 2, 2, 1)$, and descendant-heavy $(0, 1, 1, 2)$, consistently with $w_I = w_{s_0} = 0$. The benchmark artifacts display a common mandatory resource unit in addition to that formal burden:

$$\widetilde{W}(L) := 1 + W(L). \quad (6.8)$$

Thus the displayed schedule vectors are $(1, 1, 1, 1)$, $(1, 2, 2, 1)$, and $(1, 1, 1, 2)$. If $\widetilde{V}^*$ and $\widetilde{J}^*$ denote the benchmark objectives written in this display convention, then

$$\widetilde{V}^*(b; \widetilde{B}) = V^*(b; \widetilde{B} - 1), \quad \widetilde{B} \geq 1, \quad (6.9)$$

$$\widetilde{J}^*(b; \lambda) = J^*(b; \lambda) - \lambda. \quad (6.10)$$

The translation shifts capacity axes by one unit and penalized-value levels by $-\lambda$. It leaves optimal libraries, switching prices, breakpoint order, and absolute burden reductions unchanged. The capacity and penalty theorems, and their Lean counterparts, apply to W , not to $\widetilde{W}$.

Complete enumeration shows that the full source compresses uniquely to $I + D$ under every schedule, keeps frontier $(2, 4)$ and closure $\{m\}$, and saves one-half, two-thirds, or two-fifths of displayed source burden $\widetilde{W}$. These are exact finite enumeration certificates, not consequences of a greedy rule. The complete source-by-source correspondence and exact burden reductions are in Online Supplement S4.

6.5 Capacity-optimal library path

The hard-capacity optimizer is evaluated at every attainable displayed burden $\widetilde{B}$. Equal active and carrier heavy jump directly from I to $I + D$. Descendant heavy passes through the tied carrier pair $I + A$ or $I + B$; the resulting capacity complements make the second value increment exceed the first. The exact paths, globally active positive switch prices, and inactive pairwise crossing candidates are tabulated in Online Supplement S4; the main text uses the combined economic-geometry figure to avoid repeating the same staircase and breakpoints.

6.6 Penalized envelope, breakpoints, and selected burden

For each schedule and belief, the displayed penalized problem exhausts the eight affine branches $V(b, L) - \lambda \widetilde{W}(L)$. Only $I + D$ and I are globally active for $\lambda > 0$. Equal active and carrier heavy switch at 1229/288 from low belief and 2089/288 from high belief. Descendant heavy doubles the relevant burden difference, so its breakpoints are 1229/576 and 2089/576. At a breakpoint both libraries are optimal; away from it the optimizer is unique after removal of redundant equal-state libraries.

6.7 Innovation duration and elasticity values

At $\beta = 1/2$, the exact local fixed-policy discount elasticities $D_\beta = \beta V_\beta / V$ are

	ℓ	h	productive channel
I	10777/2070	14089/2070	generative
$I + A$ or $I + B$	56/15	53/15	generative
$I + D$	25/21	29/33	operational.

The zero complementary channel is reported as a zero contribution, not with an elasticity. In the registered bridge calculation with gross descendant value one, duration three, and normalized positive margin 1/16, the exact discount/survival, admission/payoff, and cost elasticities are respectively 48, 16, and -15 . These are fixed-exposure values; they do not differentiate through a library or action switch.

6.8 Floating-Point Verification

After fixing the exact solution and optimizer correspondences, the same raw-law model's `Float64` value iteration stops after 42 steps. Its final increment is 5.5422×10^{-13} , its Bellman residual is 2.7711×10^{-13} , and its `Float64` a-posteriori contraction bound is 5.5422×10^{-13} . Converting the binary `Float64` iterate exactly to rationals gives maximum error 5.5482×10^{-13} against the exact rational table, below the exactly re-evaluated residual-based certificate 5.5489×10^{-13} . The six actions match Table 1. Appendix D reports the stopping rule and compact certificate, while Online Supplement S4 reports the complete history; the policy-map response remains with the controlled diagnostics in Online Supplement S5.

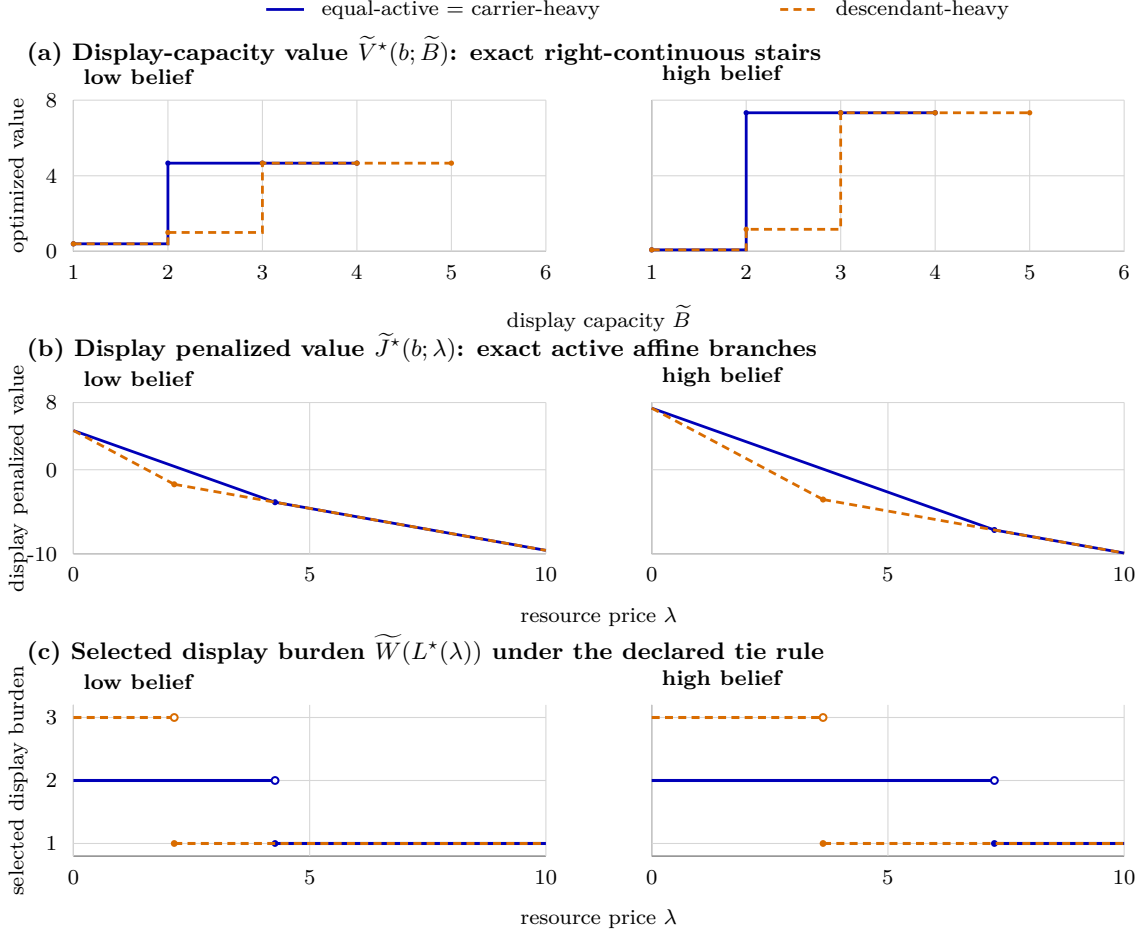


Figure 3: Economic geometry of the canonical outer optimization. Panel (a) plots $\tilde{V}^*(b; \tilde{B})$ at and between attainable exact displayed burdens as right-continuous stairs. Panel (b) plots the exact active affine branches of $\tilde{J}^*(b; \lambda)$ over the displayed range. Panel (c) plots displayed burden $\tilde{W}$ under the declared right-continuous minimum-burden tie rule; open endpoints are excluded and filled endpoints are selected. Here $\tilde{W} = 1 + W$ includes the common mandatory display unit; the formal model and Lean theorem use $w_{s_0} = 0$. Equal-active and carrier-heavy schedules coincide. Source values are exact `Rational{BigInt}`; only TeX mark placement uses rendered floating-point coordinates.

7 Controlled experiments

The experiments separate three evidentiary roles. Exact rational fixtures exercise individual optimization mechanisms; a registered randomized study summarizes how those mechanisms behave across finite generated libraries; and the two financial audits ask retrospectively how much resource an exact frontier–closure constraint costs in two larger libraries. The first two layers are finite computational evidence; the financial information boundary is stated in Appendix E. Full protocols and diagnostic tables are in Online Supplements S5–S7.

7.1 Exact optimization fixtures

The smallest exact fixtures are reported where their mechanisms are used: the local–global compression witness in Section 4, the sharp bridge loss in Section 4.5, and the capacity and price geometry in Section 6. Online Supplement S4 gives the complete canonical resource correspondences, and Online Supplement S5 retains the broader mechanism suite. All fixture values use exact rational arithmetic and complete finite enumeration.

7.2 Randomized optimal-compression study

The registered $N = 1024$ extension evaluates the resource problem with exact rationals on the frozen finite-library registry. The resulting summary is in Table 2. The registered rechecked deletion order’s zero gap is a design outcome, not a greedy-optimality theorem.

Registered estimand (unit)	Exact result or explicitly rendered mean
Exact safe-compression reduction (share of source)	Mean burden removed $\frac{31}{160}$; mean cardinality removed $\frac{1}{8}$
Registered deletion-order gap (burden units)	Mean and maximum 0; positive in 0/1024 trials
Frontier-only loss (normalized value units)	Positive in 398/1024; mean ≈ 0.03514 , conditional-positive mean ≈ 0.09041 , median 0, maximum $\frac{84963}{222163}$
Capacity value at 75% source burden (normalized value units)	Across-trial mean ≈ 0.9941
Capacity nonconcavity (trial count)	97/1024 on the registered grid; 921/1024 over all attainable burdens
Resource-price breakpoints (count per trial)	Mean total $\frac{1043}{256} \approx 4.074$; mean strictly positive $\frac{787}{256} \approx 3.074$

Table 2: Randomized optimal-compression outcomes. Fractions and integer counts are exact. A leading $\approx$ marks a decimal rendered from an exact across-trial rational mean; the committed source tables retain the full fractions and every optimizer tie. Frequencies describe the registered finite generator, not a real-library population.

Capacity rises quickly but not concavely: on average, three quarters of source burden recovers more than 99% of attainable productive value, yet many attainable staircases contain an increasing increment. Frontier-only compression is more aggressive precisely because it may discard the descendant channel; its zero median normalized loss coexists with a sharp upper tail.

The registered price intervals also separate resource demand from productive value and its channels. Prefix stability, Wilson intervals, Monte Carlo standard errors, full factorial slices, and the frozen-pilot comparison are confined to Online Supplement S5.

Price arc	Demand elasticity	Productive-value elasticity	Operational contribution	Generative contribution
$\frac{1}{4} \rightarrow \frac{1}{2}$	≈ -0.13933	≈ -0.02384	≈ -0.02651	≈ 0.00266
$\frac{1}{2} \rightarrow 1$	≈ -0.31255	≈ -0.09022	≈ -0.09193	≈ 0.00172
$1 \rightarrow 2$	≈ -0.35327	≈ -0.15161	≈ -0.13938	≈ -0.01223
$2 \rightarrow 4$	≈ -0.19792	≈ -0.11306	≈ -0.09668	≈ -0.01639

Table 3: Across-trial mean arc elasticities and channel contributions; every entry is dimensionless. Exact `Rational{BigInt}` observations are rounded to five decimal places for display. Every arc is defined in all 1,024 trials, so no undefined elasticity is imputed as zero. The two channel contributions add exactly to productive-value elasticity before rendering.

7.3 Cross-audit financial resource optimization

The two licensed-data exercises are complementary retrospective protocols with different decision timing and held-out outcome units. The terminal financial audit makes one fixed holdout evaluation after development and validation; the annual walk-forward financial audit makes five target-year decisions using trailing information through the preceding year. Neither protocol supersedes the other, and their held-out quality levels are not pooled. Both compare a current stepwise safe library with HiGHS-reported minimum-cardinality and minimum-validation-resource candidates. The terminal audit starts from 80 strategies, 38 modules, and validation-computation burden 1558; the annual walk-forward audit starts from 202 strategies, 113 modules, and burden 3508. Validation frontiers and structural closure are available at compression time. Held-out descendant quality is read only after each decision and is used solely as an ex post opportunity diagnostic; neither pruning rule tests candidate-set equality. Appendix E gives the complete information set and solver qualifications; Online Supplement S6 gives the formulas and secondary tests.

Audit	Source ($ L , W_v$)	Stepwise safe ($ L , W_v$)	HiGHS min-card. $ L $	HiGHS min- W_v ($ L , W_v$)	Frontier-only ($ L , W_v$)
Terminal	(80, 1558)	(25, 450)	25	(25, 338)	(3, 64)
Annual	(202, 3508)	(100, 1566)	100	(100, 1170)	(5, 44)

Table 4: Cross-audit financial resource optimization. $|L|$ is a strategy count and W_v is measured in prespecified validation-computation units. “HiGHS” means the mixed-integer solver reported `OPTIMAL`; it does not mean exact global optimality was established by enumeration. Selected libraries, burdens, frontiers, and closures are rechecked exactly.

HiGHS reports that the stepwise endpoints already match its minimum-cardinality candidates: both retain 25 and 100 strategies, respectively, although distinct HiGHS-reported minimizing libraries can use different identities. The HiGHS-reported minimum-weight safe candidates keep those cardinalities but lower burden from 450 to 338 and from 1566 to 1170. Relative to the source, the savings are 1220 units (78.31%) and 2338 units (66.65%); relative to the current stepwise endpoints, the HiGHS-reported weight candidates save a further 24.89% and 25.29%. Every safe library in the comparison preserves the complete frontier and all 38 or 113 source modules, and its ex post enabled-descendant opportunity quality is unchanged.

Frontier-only compression retains just 3 and 5 strategies and only 12/38 and 14/113 modules. Its ex post opportunity diagnostic falls by 0.0016 and 0.1020 in the audits’ distinct units. The resulting safety-compression trade-off is stark: minimum-cardinality safe compression removes 68.75% and 50.50% of strategies while preserving closure; frontier-only compression removes 96.25% and 97.52%, but breaks closure and loses the best held-out enabled opportunity in both audits.

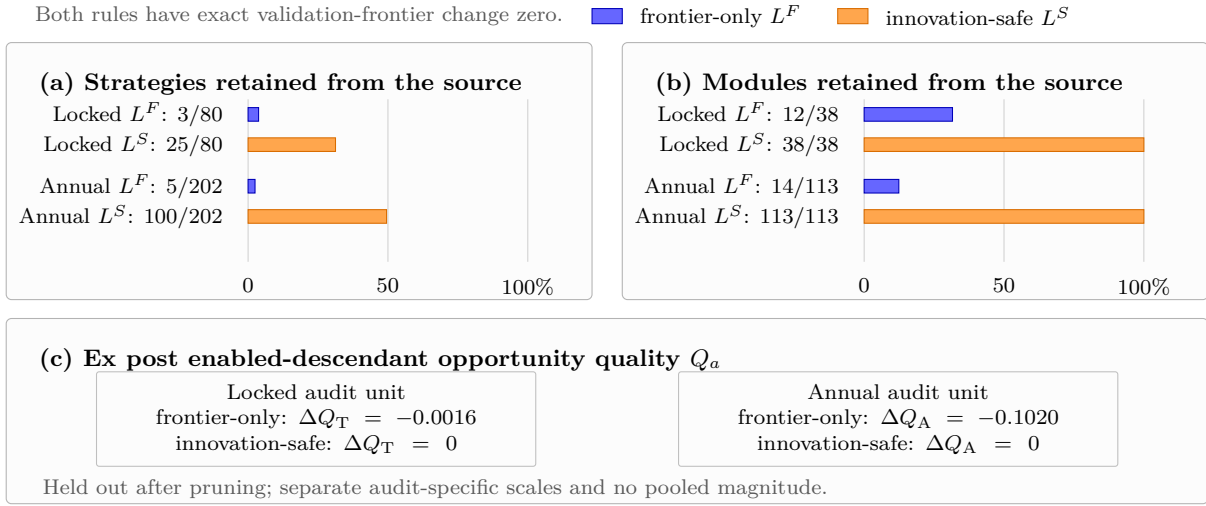


Figure 4: Financial safety–compression trade-off for the current stepwise endpoints. Panels (a)–(b) report exact retained fractions of each fixed source library and module closure. Panel (c) reports $\Delta Q_a = Q_a(L') - Q_a(L)$, the held-out change in the best enabled descendant’s audit-specific opportunity quality; the two audit units are not pooled. Both rules preserve the validation frontier, but only the safe rule preserves closure and Q_a . The displayed Q_a values are rendered audit decimals and are not pruning inputs. Table 4 adds the HiGHS-reported minimum-cardinality and minimum-weight comparisons.

The secondary coverage-ranking stress test is adverse in the terminal audit and mixed in the five annual targets. Coverage scores are post-compression ranking inputs, never pruning inputs. The terminal coverage-ranked set has locked utility -0.1560 , versus -0.0685 for the strongest comparator, with Spearman association -0.0982 . Annual marginal coverage averages 0.0704 , versus 0.0119 , but is positive in only three of five targets; mean Spearman association is 0.1279 and greedy-oracle regret is 0.315 . These retrospective quantities neither identify expected return nor establish market alpha. The secondary figure and full estimates are online.

8 Conclusion

Strategy libraries grow because research keeps producing new policies, modules, and recombinations worth retaining. That growth is productive, but it is not free: storage, validation, monitoring, governance, and operational attention are scarce resources. The economic difficulty is that a strategy is both an operational asset and a generative asset. It can improve the action frontier available today, and it can supply modules that create descendants unavailable from the current frontier alone. A compression rule that recognizes only the first role can therefore discard the source of future invention.

This paper formulates compression as a finite combinatorial optimization problem. Given a current library and additive resource weights, exact safe compression minimizes retained burden while preserving both the operational frontier and the descendant-generating closure. The projection theorem makes these two objects the sufficient productive state and hence the correct exact constraint: every safe-feasible library has the same productive value even when its resource burden differs. The innovation-safe deletion certificate is a useful local feasibility test. Rechecking that certificate after each deletion produces a certified safe endpoint, but stepwise pruning need not solve the global minimum-resource problem. Inclusion-wise irreducibility and global optimality are distinct notions, so deletion order and a complete search over feasible sublibraries can matter.

Preserving closure is economically substantive rather than a formal refinement. Frontier-only compression may leave all immediately available actions unchanged while severing the modules needed for a valuable descendant. In the sharp bridge construction, dropping closure destroys all attainable descendant value, so normalized bridge loss reaches one. Static coverage or current payoff equivalence is consequently not a sufficient safety criterion when retained strategies participate in later research.

Scarcity also determines which imperfectly retained library is optimal. A hard capacity constraint expands the feasible set only when the budget can support another relevant combination, while a resource penalty selects the upper envelope of finitely many value–burden lines. Capacity changes and resource-price changes therefore alter the optimal retained library at discrete breakpoints. The resulting selections need not be nested. Moreover, generative modules can be complements: a module may be unproductive alone but valuable with its bridge partners. That complementarity can make the value of capacity nonconcave even though capacity value remains monotone.

The value decomposition supplies the second step in the paper’s hierarchy: it interprets why the compression constraints matter. Operational value rewards actions already available, while generative value rewards future descendants and is exposed to the delay before innovation arrives. The core elasticity, duration, and switching results are in Appendix C; extended belief-occupation and interaction results are in Online Supplements S1–S3. Innovation duration measures timing exposure and channel elasticities are used only when their value denominator is positive; contribution or finite-change measures replace them at zero and at switching thresholds.

The exact fixtures, randomized study, and financial audits illustrate these optimization mechanisms at different evidentiary levels. In particular, the financial comparisons between a stepwise safe library, HiGHS-reported minimum-resource candidates, and a frontier-only library expose the safety–compression trade-off. Held-out outcomes are retrospective opportunity diagnostics only: they make no causal, forecasting, or alpha claim. Natural extensions are approximate compression with an explicit safety tolerance, endogenous information processing, continuous belief spaces, and endogenous verification of newly generated strategies. Each would enlarge the choice set while preserving the paper’s organizing question: how much resource burden can be removed without sacrificing the operational and inventive capacity of the library?

Data and Code Availability

The project repository provides the formal and computational source code together with the synthetic and exact experiment materials. It is available at

<https://github.com/dramirez37/financial-strategy-library-compression>

The financial audits use licensed CRSP/WRDS source data; raw and row-level licensed data cannot be redistributed. Users with independent licensed access can reproduce the licensed-data portions using the supplied procedures. Online Supplement S7 provides reproducibility commands and artifact records.

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A Proofs of Structural and Value Results

This appendix gives the proof support for the structural projection, safe deletion, value decomposition, and descendant-event bound. Throughout, expectations are conditional on the displayed initial state and action, and the joint completion law remains unfactorized unless a result explicitly imposes a product specialization. Finiteness is used for finite sums, attained maxima, termination, and finite quotients. Observable closure detectability is used only for the converse from dynamic observations to closure equality; contraction is used only for stationary fixed points, policy evaluation, and stationary action comparisons.

A.1 Auxiliary finite-model facts

For a finite carrier A , every probability law in the model is a table

$$\Delta_{\mathbb{Q}}(A) := \left\{ p : A \rightarrow \mathbb{Q} : p(a) \geq 0 \text{ for every } a, \sum_{a \in A} p(a) = 1 \right\}.$$

Thus beliefs, belief-kernel rows, and generation rows are normalized exact rational tables. At fixed (q, b, C) , admission assigns verified mass to accepted candidates and all rejected mass to the null outcome:

$$\Gamma(q, b, C)(\text{some}(s)) = G(q, b, C)(\text{some}(s))\nu(q, b, C, s), \quad (\text{A.1})$$

$$\Gamma(q, b, C)(\text{none}) = G(q, b, C)(\text{none}) + \sum_{s \in S} G(q, b, C)(\text{some}(s))(1 - \nu(q, b, C, s)). \quad (\text{A.2})$$

Nonnegativity is immediate, and summing these rows recovers the unit mass of G , so $\Gamma(q, b, C)$ is normalized.

Conditional on the fixed initial belief b , state (F, C) , and project q , the joint completion law on $\mathbf{b} = (b_0, \dots, b_{d_q})$ and admitted outcome o has marginals

$$\sum_o \Lambda_q(\mathbf{b}, o \mid b; F, C) = \mathbf{1}\{b_0 = b\} \prod_{t=0}^{d_q-1} P_B(b_t, b_{t+1}), \quad (\text{A.3})$$

$$\sum_{\mathbf{b}} \Lambda_q(\mathbf{b}, o \mid b; F, C) = \Gamma(q, b, C)(o). \quad (\text{A.4})$$

The raw terminal law is therefore

$$\mathcal{Q}_q^{\text{raw}}(b', L' \mid b, L) = \sum_{\substack{\mathbf{b}, o: b_{d_q} = b' \\ L \oplus o = L'}} \Lambda_q(\mathbf{b}, o \mid b; F_L, C_L).$$

For realizable $K = (F, C)$, deterministic pushforward gives

$$\overline{T}_q(K' \mid b, K) := \sum_{o: \text{addK}(K, o) = K'} \Gamma(q, b, C)(o), \quad (\text{A.5})$$

$$\overline{\mathcal{Q}}_q(b', K' \mid b, K) := \sum_{\substack{\mathbf{b}, o: b_{d_q} = b' \\ \text{addK}(K, o) = K'}} \Lambda_q(\mathbf{b}, o \mid b, K). \quad (\text{A.6})$$

These pushforwards preserve nonnegativity and total mass and are supported on realizable compressed states by (3.7).

Lemma A.1 (Finite-library frontier and closure calculus). *For every raw library L and belief b ,*

$$j_s(b) \leq F_L(b) \quad (s \in L), \quad F_L(b) \geq 0, \quad \exists s^* \in L : j_{s^*}(b) = F_L(b).$$

If $L \subseteq L'$, then $F_L(b) \leq F_{L'}(b)$ for every b and $C_L \subseteq C_{L'}$.

Proof. The inactive strategy makes L nonempty and gives the nonnegative row. The finite maximum defining $F_L(b)$ is attained and bounds every represented profile. Library inclusion enlarges the maximizing set and the raw module union; closure monotonicity gives the closure inclusion. $\square$

A belief is reachable at date t exactly when its transition mass is positive; finite occupation is the date-by-date mass weighted by the relevant discount and survival powers. On a finite linear order, every nonempty upper set has a least member b^* and equals $\{b : b^* \leq b\}$. These two facts supply the occupation and cutoff conventions used in the supporting coverage results.

A.2 Projection and dynamic equivalence

Proof of Theorem 3.2. Fix an initial belief b , raw library L , and feasible project q . The admitted law is normalized by (A.1)– (A.2); pushing the joint completion law through the deterministic raw update gives the normalized laws (A.5)– (A.6). The pointwise identity

$$K_{L \oplus o} = \text{addK}(K_L, o)$$

shows that the next compressed state is realizable and that its law depends on L only through K_L . Libraries in the same fiber therefore have identical project availability, cost, duration, incumbent-reward blocks, and joint terminal belief–compressed-state laws. This proves the controlled semi-Markov projection at decision epochs; an unfinished project additionally records its identity and remaining duration.

For finite-horizon values, compare corresponding actions. Continue reduces the horizon by one and each research action by $d_q > 0$, so strong induction identifies every raw and compressed action value and then their attained maxima. For the stationary clause, Bellman intertwining maps a compressed fixed point to a raw fixed point; discounted contraction supplies fixed-point and policy-evaluation uniqueness. A maximizing compressed selector exists by finite attainment and lifts through the same action-value identity. $\square$

Proof of Proposition 6.2. Fix a decision state and a remaining finite calendar horizon. The feasible action set consists of Continue and every available project whose positive duration fits the remaining horizon. The project and state carriers are finite, so this action set is finite and its maximum is attained. This proves finite-horizon attainment without contraction.

For the stationary claim, cast the exact rational probabilities and payoffs to reals and condition each expectation on the current state and chosen action. Expectation under either the belief kernel or the joint completion law is nonexpansive in the finite-product sup norm. No independence between the belief path and admitted outcome is needed. Continue has modulus β . A positive-duration project has modulus $\beta^{d(q)} \leq \beta$; its cost and incumbent operating block cancel when two continuation tables are compared. Maximization over the finite feasible action set preserves this common pointwise bound, so both raw and compressed Bellman operators are contractions.

This is the only place in the argument where contraction is used. Banach’s theorem gives a unique fixed point for each Bellman operator, uniform value- iteration convergence, and (6.6). Bellman intertwining maps the compressed fixed point to a raw fixed point; fixed-point uniqueness gives raw/compressed value equality. Finiteness supplies at least one maximizing stationary action

at each state, defining a selector π^* ; the selector itself need not be unique. Its fixed-action operator obeys the same contraction bound, and $T_{\pi^*}V_\infty = V_\infty$. Uniqueness of policy evaluation then yields $V^{\pi^*} = V_\infty$, and the projection identity lifts the selector to the raw state space. Detectability is not used. $\square$

Proof of the characterization clause of Theorem 4.5. If $K_L = K_{L'}$, raw factorization identifies generation, verification, availability, cost, duration, incumbent rewards, and the conditional joint completion law. The admitted laws and compressed pushforwards agree, giving all five observations in (4.5). Conversely, dynamic innovation equivalence gives frontier equality. If the closures differed, observable closure detectability would supply a project with a different availability-tagged cost, duration, or joint terminal observation, a contradiction. Thus the closures and compressed states agree. $\square$

The five componentwise equalities in (4.5) make $\sim_{\text{DI}}$ an equivalence relation. The proof of Theorem 3.2 applies verbatim to any pair $L \sim_{\text{DI}} L'$: finite-horizon and stationary values agree, the finite raw-library carrier has a finite quotient, and value is well defined on each class. Equality of compressed states implies this equivalence by the forward characterization just proved. Finally, if a representation r has the property that $r(L) = r(L')$ forces equality of all five availability-tagged observations, then equality of r implies $L \sim_{\text{DI}} L'$. This restricted refinement is definitional; it is not a generic coarsest-bisimulation or full-abstraction result.

A.3 Safe deletion and rechecking

Proof of the deletion clauses of Theorem 4.5. For an active $s \in L$, equation (4.6) is componentwise equality of $K_{L-s} = (F_{L-s}, C_{L-s})$ and $K_L = (F_L, C_L)$. The two redundancy equalities therefore imply dynamic equivalence and finite-horizon value preservation by the forward projection argument. Under detectability, the converse characterization in A.2 makes them necessary for dynamic innovation equivalence. Compressed-state equality also preserves the stationary value and complete action comparison under the discounted contraction. $\square$

Proof of Corollary 4.6. At each accepted deletion, apply the sufficient direction to the current library, then compose the resulting compressed-state equalities along the trace. Every endpoint remains source-feasible and has the source's productive values. Each step removes an active strategy, so the procedure terminates. If a complete final scan finds no certified deletion, the endpoint is one-deletion irreducible. The trace certifies feasibility and value preservation, not minimum resource burden. $\square$

A.4 Operational/generative value and descendant bound

Proof of Theorem 5.2. Substitute $\Omega_n = U_n - P_n$ into the definition of Δ_n^{gen} . The two passive-value differences combine into Δ_n^{op} , and the remaining full-value difference is $\mathcal{I}_n$. This is the pointwise identity (5.5) at fixed (n, b, L, s) . $\square$

Proof of Corollary 5.3. Equal frontiers make the passive recursions identical. If closure is also unchanged, the two realizable compressed states coincide, so Theorem 3.2 identifies their full values. $\square$

Proposition A.2 (Research-premium monotonicity under project dominance). *Let L^- and L^+ have the same frontier and $C_{L^-} \subseteq C_{L^+}$. Suppose every old project remains feasible and its complete*

action value—including cost, duration, operating rewards, completion law, and continuation value—is weakly higher at L^+ . Then

$$\Omega_n(b, L^-) \leq \Omega_n(b, L^+)$$

for every finite horizon and belief. Hence a frontier-silent insertion satisfying this project-dominance certificate has nonnegative generative insertion value.

Proof. Induct on the finite horizon. Continue has the same passive reward because the frontiers agree. Every old research action remains feasible and has a weakly higher complete action value by hypothesis, while closure enrichment may add actions. Maximization therefore weakly raises full value. Subtracting the common passive value gives the premium order. $\square$

Proof of Theorem 5.5. Condition on $(B_0, K_0, q) = (b, K_{L^+}, q)$. Full value at L^+ dominates both Continue and commitment to the enabled project q . Subtracting passive value from the commitment value leaves initiation cost, the exact operating adjustment $A_{q,h}^{\text{op}}$, and discounted continuation gain. Insertion-only updates and retention of the passive policy make every omitted outcome's continuation contribution nonnegative. On the distinguished event the gain is at least $G(B_d)$, whose contribution under the joint completion law is

$$\sum_{b'} \eta_{q,g}(b' \mid b, K_{L^+}) G(b').$$

Frontier silence removes operational insertion value, and the zero-premium deleted comparator converts the retained premium into generative insertion value. Maximizing research against Continue yields (5.9); setting $A^{\text{op}} = 0$ gives (5.10). Under the product specialization, substitution of $\eta_{q,g} = \pi \rho^d \mu_{q,d}$ gives $\pi \rho^d \bar{G}_{q,d}(b)$. The sign comparisons in Corollary 5.6 follow termwise from nonnegative joint mass and gain, subject to its stated qualification when duration changes other terms. $\square$

Example A.3 (Exact one-belief retained carrier). Let $d = 1$, $\beta = 1/2$, $\kappa = 0$, and $\pi = \rho = 1$. A zero-profile bridge uniquely enables a descendant with gain $G = 2$; deletion leaves no research project. Operation continues against a zero frontier, so $A^{\text{op}} = 0$, and the bound gives

$$1 \leq \Delta_2^{\text{gen}}.$$

The extended gap-sum, coverage, and comparative-static arguments are in Online Supplements S1 and S3.

A.5 Essential assumption-boundary examples

These are the two structural failures needed to delimit the headline projection and characterization results. Auxiliary finite-model examples are in Online Supplement S3.

Counterexample A.4 (Raw identifiers defeat frontier-closure sufficiency). There are exact finite libraries with the same frontier and closure but different candidate laws when a broader generator is allowed to inspect membership of a hidden raw strategy identifier. With unit admission the hidden identifier generates a module-bearing candidate, so the exact next-compressed-state laws also differ. That generator fails the raw frontier-closure factorization imposed by the unified raw model.

Counterexample A.5 (Behaviorally invisible raw closure). There is an exact finite raw process whose project menu is empty at every state. Two libraries have the same frontier and different closures but agree on all five unified dynamic-innovation observations. The process fails raw closure detectability, so the characterization's converse cannot omit that assumption.

B Proofs of Resource-Optimization Results

This appendix supplies the finite arguments behind the resource results in Section 4. All comparisons hold at the fixed belief, primitive model, and eligible catalog displayed with the corresponding result.

B.1 Minimum-resource representation and local/global boundary

Proof of Theorem 4.1. The source L belongs to $\mathcal{P}_{\text{safe}}(L)$, so the feasible family is nonempty. It is a subset of the finite powerset of L , hence the rational burden attains a minimum. Every feasible L' satisfies $K_{L'} = K_L$, so the forward projection theorem identifies all productive finite-horizon values. Bellman intertwining and discounted contraction give the stationary value and complete optimal-action correspondence. $\square$

Proof of Theorem 4.9, forward implication. If a global minimum retained a safely deletable active strategy s , then positivity of w_s would give

$$W(L' \setminus \{s\}) = W(L') - w_s < W(L'),$$

contradicting minimality. Thus every global minimum is one-deletion irreducible. A rechecked safe-deletion trace is feasible because its exact compressed-state equalities compose, but composition alone is not an optimization argument. $\square$

The converse fails in the exact $(2, 2, 3)$ construction. Under identity closure and the common zero frontier, a retained active set is safe exactly when it covers both m_1 and m_2 . The singleton carriers s_1, s_2 have weights 2, 2, and their bundle s_{12} has weight 3. The five safe active sets are

$$\{s_1, s_2\}, \quad \{s_{12}\}, \quad \{s_1, s_{12}\}, \quad \{s_2, s_{12}\}, \quad \{s_1, s_2, s_{12}\},$$

with respective burdens 4, 3, 5, 5, 7. Deleting the unique heaviest safe strategy first leaves the two singleton carriers and burden four, while the bundle-only library is the unique global minimum with burden three. Both endpoints preserve the source frontier and closure; the failure is global resource optimality, not productive value.

B.2 Sharp frontier-only loss

Proof of Theorem 4.11. Use the canonical bridge catalog of Section 4: the retained library has frontier $F_L = (3, 3)$, closure $C_L = \{m\}$, and a dominated unique carrier of m . Deleting that carrier preserves the frontier but makes the sole descendant mass zero. Conditional on retention and commitment to the unique project, sequential generation and admission give admitted descendant mass $\rho^d \pi$. Terminal reward R arrives after d dates, so the retained one-project envelope is

$$\max\{0, \beta^d \rho^d \pi R - \kappa\},$$

whereas the frontier-only comparator is zero. The incumbent operating block cancels under frontier equality. At $R = C$, the worthwhile condition selects the research branch and yields

$$\text{Loss}_d = \beta^d \rho^d \pi C - \kappa.$$

For $0 \leq R \leq C$, monotonicity of the two-branch envelope gives the same quantity as an upper bound, and $R = C$ attains it. Thus the bound is sharp within the stated canonical comparison. Substituting $(d, \beta, \rho, \pi, \kappa, C) = (1, 1/2, 1, 1, 0, 2M)$ gives Corollary 4.12. $\square$

B.3 Capacity-constrained value

Proof of Theorem 4.13. The inactive-only library makes every nonnegative budget feasible. The eligible family is finite, so each feasible subfamily has a maximizing library. If $B_1 \leq B_2$, the first feasible family is contained in the second, proving monotonicity. Order the finite set of attainable burdens. Between consecutive burdens the feasible family is unchanged, hence its maximum is unchanged; every strict value change therefore occurs at an attainable burden, and every exact forward difference is nonnegative. $\square$

Increasing increments remain possible. In the exact two-unit joint-capability example,

$$V^*(0) = 0, \quad V^*(1) = 0, \quad V^*(2) = 1,$$

so the successive unit increments are 0 and 1. The first unit cannot assemble the required pair, while the second completes it. This also shows why capacity value need not be concave; separately, the discrete price problem may omit an attainable capacity optimum.

B.4 Penalized finite envelope

Proof of Theorem 4.14. At fixed (θ, b) , each eligible library supplies the affine branch

$$J_{L'}(\lambda) = V_\theta(b, L') - \lambda W(L').$$

The finite maximum is attained and is continuous, convex, and piecewise affine, with changes of maximizing face only at finitely many pairwise intersections. Nonnegative burdens make every branch, and therefore the envelope, nonincreasing.

Let L_1 optimize at λ_1 and L_2 at $\lambda_2 > \lambda_1$. Optimality gives

$$\begin{aligned} V_1 - \lambda_1 W_1 &\geq V_2 - \lambda_1 W_2, \\ V_2 - \lambda_2 W_2 &\geq V_1 - \lambda_2 W_1. \end{aligned}$$

Adding yields $(\lambda_2 - \lambda_1)(W_1 - W_2) \geq 0$, hence $W_2 \leq W_1$. This applies to every optimizer pair and proves the set-valued burden order. Strict dominance on a neighborhood identifies one affine branch and its slope $-W(L')$; no derivative is selected at a tie, and the proof imposes no nesting of raw libraries. $\square$

The supporting conditional replacement formulation is in Online Supplement S3.

C Supporting sensitivity results

This appendix retains only the sensitivity results needed to interpret the main paper’s value and resource-optimization statements. Each calculation holds along the named parameter path or conditional on the displayed initial belief and fixed action. Unnamed primitives are held fixed. Finiteness is used for termwise differentiation and finite action maximization. No closure-detectability or stationary-contraction argument is used here.

The extended belief-occupation, coverage, frontier–closure interaction, and counterexample material is in Online Supplements S1–S3.

C.1 Bridge-margin sensitivity

The canonical bridge isolates the positive-value interior of the frontier-only loss in the main paper’s sharp normalized relaxation result. Write

$$A_{\text{br}} := \beta^d \rho^d \pi C, \quad M_{\text{br}} := A_{\text{br}} - \kappa, \quad m_{\text{br}} := \frac{M_{\text{br}}}{A_{\text{br}}}. \quad (\text{C.1})$$

On the declared real extension with fixed integer d , positive gross coordinates, and $M_{\text{br}} > 0$, realized bridge value equals the margin. Its named-coordinate elasticities are

$$(\varepsilon_{\beta}^M, \varepsilon_{\rho}^M, \varepsilon_{\pi}^M, \varepsilon_C^M, \varepsilon_{\kappa}^M) = \left(\frac{d}{m_{\text{br}}}, \frac{d}{m_{\text{br}}}, \frac{1}{m_{\text{br}}}, \frac{1}{m_{\text{br}}}, -\frac{1 - m_{\text{br}}}{m_{\text{br}}} \right). \quad (\text{C.2})$$

Threshold fragility is governed by the normalized positive margin m_{br} . The magnitudes diverge as $m_{\text{br}} \downarrow 0$, but not merely because $M_{\text{br}} \downarrow 0$: if cost is zero and gross scale vanishes, then $m_{\text{br}} = 1$. At $M_{\text{br}} = 0$, bridge-value elasticity is undefined. At and below that threshold we report the signed margin, the action region, or exact finite changes in realized value, never a percentage elasticity.

For the derivation, condition on the canonical project and hold every unnamed coordinate fixed. Differentiating on the positive real extension gives

$$\beta \partial_{\beta} M_{\text{br}} = d A_{\text{br}}, \quad \rho \partial_{\rho} M_{\text{br}} = d A_{\text{br}}, \quad \pi \partial_{\pi} M_{\text{br}} = C \partial_C M_{\text{br}} = A_{\text{br}}, \quad \kappa \partial_{\kappa} M_{\text{br}} = -\kappa.$$

On $M_{\text{br}} > 0$, realized value $[M_{\text{br}}]_{+}$ equals the signed margin. Dividing by M_{br} and using $m_{\text{br}} = M_{\text{br}}/A_{\text{br}}$ yields (C.2). At the threshold the positive-part value has no ordinary percentage elasticity. The costless path $\kappa = 0$, $A_{\text{br}} \downarrow 0$ keeps $m_{\text{br}} = 1$, which is why vanishing net level alone does not imply the threshold-fragility limit.

C.2 Operational/generative elasticity contributions

Let one named positive scalar x vary the same primitive in total insertion value $I(x)$, operational value $O(x)$, and generative value $G(x)$, holding every unlisted primitive fixed and remaining on a differentiable branch. Differentiating $I = O + G$ gives the denominator-free identity

$$xI'(x) = xO'(x) + xG'(x).$$

Proposition C.1 (Operational–generative elasticity contributions). *At $x_0 > 0$, suppose $I(x) = O(x) + G(x)$ throughout a neighborhood of x_0 , and the three channel paths are differentiable there along the same named perturbation. If $I(x_0) > 0$, define*

$$C_x^{\text{op}} := \frac{x_0 O'(x_0)}{I(x_0)}, \quad C_x^{\text{gen}} := \frac{x_0 G'(x_0)}{I(x_0)}.$$

Then

$$\varepsilon_x^I = C_x^{\text{op}} + C_x^{\text{gen}}.$$

Only when both channel levels are positive may this be written as

$$\varepsilon_x^I = \frac{O}{I} \varepsilon_x^{\text{op}} + \frac{G}{I} \varepsilon_x^{\text{gen}}.$$

The contribution form remains meaningful when one channel level is zero; the corresponding ordinary component log elasticity does not. If total value is zero, no total elasticity or total-normalized contribution is reported. The scaled derivative identity above, or the exact finite-change identity $\Delta I = \Delta O + \Delta G$, remains well defined.

Along one fixed differentiable scalar path, differentiating the level identity and applying common scaling gives

$$xI'(x) = xO'(x) + xG'(x).$$

If $I(x) > 0$, division by the total level gives the two signed contributions in Proposition C.1. Rewriting a contribution as

$$\frac{xO'}{I} = \frac{O}{I} \frac{xO'}{O}$$

requires $O > 0$, and the analogous rewrite requires $G > 0$. If a component vanishes, its scaled derivative remains available but its log elasticity does not. If $I = 0$, neither total elasticity nor a total-normalized contribution is defined; the level derivative or the exact identity $\Delta I = \Delta O + \Delta G$ is the valid replacement.

Belief	Library	V	V^{op}	V^{gen}	D_β	E_β^{op}	E_β^{gen}
low	I	$\frac{115}{288}$	0	$\frac{115}{288}$	$\frac{10777}{2070}$	—	$\frac{10777}{2070}$
low	I+A	1	0	1	$\frac{56}{15}$	—	$\frac{56}{15}$
low	I+D	$\frac{14}{3}$	$\frac{14}{3}$	0	$\frac{25}{21}$	—	—
high	I	$\frac{23}{288}$	0	$\frac{23}{288}$	$\frac{14089}{2070}$	—	$\frac{14089}{2070}$
high	I+A	$\frac{7}{6}$	0	$\frac{7}{6}$	$\frac{53}{15}$	—	$\frac{53}{15}$
high	I+D	$\frac{22}{3}$	$\frac{22}{3}$	0	$\frac{29}{33}$	—	—

Table 5: Exact productive value, channel levels, and fixed-policy discount elasticity in the canonical benchmark. The D_β column is not an application of the finite fixed-exposure duration theorem. A dash explicitly marks an undefined ordinary component elasticity at a zero component level; it is never recorded as zero. Scaled contributions are evaluated directly.

C.3 Fixed-exposure innovation duration

For a fixed nonnegative, nonzero exposure sequence $z_0, \dots, z_{H-1}$ and effective discount $\alpha > 0$, let

$$\Psi_H(\alpha) := \sum_{t=0}^{H-1} \alpha^t z_t, \quad \omega_t := \frac{\alpha^t z_t}{\Psi_H(\alpha)}, \quad D_\Psi := \sum_t t \omega_t, \quad C_\Psi := \sum_t \omega_t (t - D_\Psi)^2. \quad (\text{C.3})$$

Because the sequence is nonzero, $\Psi_H(\alpha) > 0$. The exact finite scaled-derivative identities are

$$\alpha \Psi'_H(\alpha) = D_\Psi \Psi_H(\alpha), \quad \alpha D'_\Psi(\alpha) = C_\Psi \geq 0. \quad (\text{C.4})$$

Equivalently, by the ordinary chain rule, D_Ψ is the derivative of $\log \Psi_H$ with respect to $\log \alpha$, and C_Ψ is the corresponding derivative of D_Ψ . These log-composition readings are interpretations of the

displayed scaled identities. Innovation duration is therefore the contribution-weighted date at which fixed uncovered value arrives, and innovation convexity is its timing variance. A higher effective discount shifts weight toward later exposures and weakly raises duration. This is not primitive project delay, time to admission, or a convexity theorem for an optimized Bellman value whose exposure sequence or selected action changes. If a candidate has zero exposure and hence zero potential, duration and log elasticity are not defined; level contributions or finite differences are used instead.

For the derivation, introduce the finite moments

$$N_1(\alpha) := \sum_t t \alpha^t z_t, \quad N_2(\alpha) := \sum_t t^2 \alpha^t z_t.$$

Termwise finite differentiation gives

$$\alpha \Psi'_H(\alpha) = N_1(\alpha), \quad \alpha N'_1(\alpha) = N_2(\alpha).$$

Since $\Psi_H > 0$, $D_\Psi = N_1/\Psi_H$, and the quotient rule yields

$$\alpha D'_\Psi = \frac{N_2 \Psi_H - N_1^2}{\Psi_H^2} = \sum_t \omega_t (t - D_\Psi)^2 = C_\Psi \geq 0.$$

These are the two identities in (C.4). The nonnegative curvature is curvature of $\log \Psi_H$ in $\log \alpha$ for a fixed exposure sequence. Changing project duration, belief dynamics, completion laws, or the selected action changes the sequence and lies outside this derivative.

C.4 Finite switch and one-sided-change conventions

Within a strictly dominant price branch, the penalized envelope inherits the branch slope $-W(L')$. A pairwise crossing is only a candidate switch; it becomes a resource-price breakpoint only when both branches are globally active. At an unequal-burden tie the optimizer is set-valued, so one-sided slopes or cross-breakpoint finite changes replace a derivative selected by an unstated tie breaker.

The same distinction guides productive-parameter diagnostics. For a finite library family, record the best-versus-second-best objective margin; for a Bellman node, record the corresponding action-value margin. A positive margin measures distance from a tie in value space. A zero margin flags a possible switch, but distinct libraries or actions may generate identical branches, so it is not sufficient for a change of optimizer. Exact resource-price breakpoints and active branches are reported in the main paper’s canonical model; adjacent-grid action-switch brackets are confined to the online supplement and registered experiment records. The registered primitive-coordinate brackets are not asserted to contain an exact root. At an actual switch with positive value, one-sided or arc elasticities may be reported; at zero value, only level slopes and finite differences are defined.

D Computational verification

D.1 Exact and floating-point certificates

The canonical solver compiles rewards and discounted transition rows from the registered raw embedded laws in `experiments/configs/unified_canonical_benchmark.toml`. Its exact mode uses `Rational{BigInt}` throughout. Exact rational policy iteration terminates with Bellman residual zero. The independently compiled `Float64` mode starts from $v_0 = 0$, forms $v_{n+1} = \mathsf{T}v_n$, and stops at the first n satisfying

$$\|v_n - v_{n-1}\|_\infty \leq \tau, \quad \tau = 10^{-12}, \quad (\text{D.1})$$

subject to the registered cap of 10,000 iterations. Failure to meet the tolerance before the cap raises an error rather than emitting a converged row.

At every iteration the solver separately records the Float64 Bellman residual

$$r_n^{64} := \|\mathsf{T}_{64}v_n - v_n\|_\infty, \quad (\text{D.2})$$

the Float64 a-priori contraction bound

$$\|v_n - V_\infty\|_\infty \leq \frac{\beta_{64}^n}{1 - \beta_{64}} \|v_1 - v_0\|_\infty, \quad (\text{D.3})$$

and the Float64 a-posteriori contraction bound

$$\|v_n - V_\infty\|_\infty \leq \frac{r_n^{64}}{1 - \beta_{64}}. \quad (\text{D.4})$$

Thus the residual is not substituted for the registered stopping rule.

The solver then applies a separate exact check to the final iterate. It converts every binary Float64 value to its exact `Rational{BigInt}` representation $\hat{v}_{42}$, evaluates $\hat{r}_{42} = \|\mathsf{T}_{\mathbb{Q}}\hat{v}_{42} - \hat{v}_{42}\|_\infty$ with the exact rational Bellman operator, and forms $\hat{r}_{42}/(1 - \beta_{\mathbb{Q}})$. This exactly re-evaluated residual-based certificate is distinct from the Float64 a-posteriori contraction bound above.

Verification item	Recorded result
Exact stationary solve	<code>Rational{BigInt}</code> ; Bellman residual 0
Float64 stopping point	$n = 42$; final increment 5.54223×10^{-13}
Final Float64 diagnostics	$r_{42}^{64} = 2.77112 \times 10^{-13}$; a-priori contraction bound 2.52953×10^{-12} ; a-posteriori contraction bound 5.54223×10^{-13}
Exactly evaluated Float64 versus exact	maximum absolute error after exact rationalization 5.5482×10^{-13} , below the exactly re-evaluated residual-based certificate 5.5489×10^{-13}
Selected actions	agreement on all six compressed state-belief pairs
Outer resource optimization	all eight canonical libraries exhaustively evaluated for every belief and registered weight schedule; all optimizer ties retained
Envelope bookkeeping	every pairwise crossing candidate recorded separately from the subset of globally active breakpoints

Table 6: Compact verification record for the canonical dynamic program and outer resource optimization.

D.2 Scope of the retained checks

The dynamic implementation checks 32 raw/compressed embedded transition rows, 144 finite-horizon values, 128 finite-horizon actions, 16 stationary values, and 16 lifted stationary actions. It independently evaluates the selected stationary policy over exact rationals and records every full-duration belief path, admission atom, and operating-reward block. The resource implementation recomputes frontier, closure, burden, productive value, and net value exactly for each enumerated library. Numerical solver status is never used as an exact certificate.

The main paper reports the economically relevant stationary solution. Online Supplement S4 contains the full 42-row convergence history and figure, horizon-by-horizon values and actions, stationary action values, path and admission atoms, safe-compression correspondences, capacity paths, penalized intervals, and all active and inactive pairwise crossing candidates.

E Experiment Design and Information Set

E.1 Registered randomized design

The registered synthetic study fixes $N = 1024$ horizon-four trials in a complete 2^7 factorial with eight replications in each of 128 cells. The master seed and all derived trial seeds were fixed before outcomes. Profiles, kernels, project laws, losses, and values use exact `Rational{BigInt}` arithmetic.

Raw realizability is a failure gate, not a post hoc filter. Each interaction observation must be generated from four realizable raw-corner libraries in one catalog and module system. Every enumerated compressed state must have a raw witness, and raw and compressed values must agree exactly at every registered belief and remaining horizon. The design also requires exact frontier, closure, operational, generative, total-loss, and decomposition agreement under innovation-safe pruning. The experiment is a synthetic structural stress test of the registered finite generator, not a population sample or a population-frequency estimate. Full estimand equations, response grids, factor slices, sequential diagnostics, and the frozen pilot are in Online Supplement S5.

E.2 Financial information set

The terminal financial audit uses one fixed development–validation–holdout split. The annual walk-forward financial audit uses five target-year decisions, each based on information through the preceding year. Their held-out outcomes and resampling units differ, so the two protocols are defined and reported separately throughout the paper and supplement.

The validation frontier is known at pruning time, and structural closure is known at the same time. Held-out descendant quality is first read only after pruning is fixed. For annual target year y , scores and selected sets use information only through $y - 1$; target-year outcomes remain held out until after the decision hash is fixed. The greedy oracle uses target-year outcomes and is therefore infeasible. No held-out quantity enters either compression rule.

Table 7: Empirical information-set classification. “Available” records the locked pipeline’s information set, not a claim that every available quantity was used.

Quantity	Classification	Timing and decision role
Current frontier	Available at pruning time	Validation-derived; frontier equality is used by both deletion rules.
Module closure	Available at pruning time	Structural; closure equality is used only by innovation-safe deletion.
Compression ratio	Retrospective descriptive statistic	Computed from final library size after pruning; outcome-blind and unused in deletion.
Enabled candidate set	Available at pruning time	Structurally derivable from catalog requirements and closure; neither deletion rule tests candidate-set equality.
Ex post enabled-descendant opportunity quality	Available only in the held-out audit	Computed from locked utility or realized annual coverage after pruning; never an algorithm input.
Coverage ranking score	Available during validation	Computed after compression from validation or pre-target trailing data; used for candidate selection, not pruning.

Quantity	Classification	Timing and decision role
Locked net utility	Available only in the held-out audit	First accessed after the terminal pruning, rankings, selections, and decision hash are fixed.
Annual realized coverage	Available only in the held-out audit	Target-year outcome first accessed after that year’s score, selection, and decision hash are fixed.
Greedy-oracle regret	Oracle comparator	Constructed from target-year realized gaps and occupation; infeasible at pruning and selection time.
Module uniqueness	Retrospective descriptive statistic	Post-compression structural summary; uses no held-out outcome and is not a deletion condition.
Descendant dependence	Retrospective descriptive statistic	Post-compression structural count of candidates disabled by a one-policy removal; uses no candidate quality and is not a deletion condition.

The table records actual pipeline use as well as logical availability. Frontier and closure are the only pruning inputs: frontier equality is common to both deletion rules, and closure equality is the additional innovation-safe gate. Coverage scores are post-compression selection inputs, whereas descendant quality, realized coverage, rank diagnostics, intervals, and oracle regret are retrospective or held-out quantities. The full code-level audit and secondary-test formulas are in Online Supplement S6.

E.3 Data and optimization qualifications

Both financial audits use a licensed CRSP/WRDS source. Raw and row-level data are nonredistributable. The ETF universes are retrospective and are not survivorship-free; neither audit is prospective.

For the reported mixed-integer searches, **OPTIMAL** is the status reported by HiGHS and is numerical global-search evidence, not an exact enumeration certificate. Every displayed selected library, integer burden, frontier, and closure is rechecked exactly after the solve. The financial audits support no causal, forecasting, alpha, or deployable-performance claim. Full financial formulas, secondary results, resampling details, and the mixed or adverse outcomes are in Online Supplement S6; configuration histories, hashes, amendments, and artifact paths are in Online Supplement S7.

F Verification and Evidence Status

Result family	Human proof	Formal verification	Exact/numerical validation	Controlled evidence	Boundary
Raw projection and sufficient state	Complete	Lean-verified exact finite encoding	Exact raw/compressed fixtures	Raw-witness stress tests	Retention burden is outside the productive state; the hidden-state expectation adapter is informal
Exact safe compression	Complete	Lean-verified finite result	Exhaustive minima and exact selected-library rechecks	Canonical, randomized, and financial audits	Stepwise irreducibility is not global minimality
Unsafe closure relaxation	Complete	Lean-verified bridge family	Exact bridge and scaled fixtures	Registered frontier-only losses	Normalized loss uses the reward cap; arbitrary additive loss scales rewards
Penalized retention envelope	Complete	Lean-verified finite envelope and burden order	Exact crossings, active breakpoints, and optimizer margins	Canonical and randomized price paths	Crossings are candidates until globally filtered
Capacity-constrained value	Complete	Lean-verified finite claims	Complete finite enumeration	Canonical and randomized capacity paths	No general diminishing returns or zero-duality-gap theorem
Value and sensitivity	Complete	Lean-verified encoded identities	Exact derivative identities and rational fixtures	Registered elasticity and duration surfaces	Chain-rule interpretations only; use one-sided or finite changes at switches
Dynamic solution	Complete	Lean-verified finite Bellman layer	Exact solution, zero residual, and optimizer/action checks	Canonical resource and robustness runs	Resource choice is an outer one-time decision
Financial information audit	Not applicable	Not applicable	Locked hashes, exact closures, and selected-library rechecks; HiGHS status remains numerical	Two retrospective audits	No causal, forecasting, alpha, or deployable-performance claim